LFB SQL Capstone Report

Oct 10, 2026 · @Mustafa Malik

1. Dataset Profile

1.1 Subject and social problem

This project analyses every incident attended by the London Fire Brigade (LFB), and every fire engine sent to those incidents, from January 2025 to June 2026.

The social problem is emergency response. How quickly a fire engine arrives affects whether people survive and how much property is lost, and response performance is not the same across London. A large share of call-outs are also false alarms, which tie up fire engines that may be needed elsewhere. The data shows where response is slowest, what causes delays, and how fire engine capacity is used against need.

1.2 Source

Both tables are published by the London Fire Brigade on the London Datastore and are updated monthly.

Table	Published dataset	What one row is
incidents	London Fire Brigade Incident Records	One incident attended by the LFB
mobilisations	London Fire Brigade Mobilisation Records	One fire engine sent to one incident

The two tables link on `incident_number`: one incident can have several mobilisations.

1.3 Size and period

The analysis covers 18 complete months, 1 January 2025 to 30 June 2026.

Table	Rows as downloaded	Rows after cleaning	Columns
incidents	358,338	209,908	39
mobilisations	336,005	312,817	24

The downloaded files covered different periods. Incidents ran from 1 January 2024 to 31 July 2026; mobilisations ran from 1 January 2025 to 30 July 2026. The period was set to the months both files cover in full:

- **2024 removed:** no mobilisation data exists for it, so response times and delays could not be analysed for that year.
- **July 2026 removed:** the mobilisations file ends on 30 July, so the month is incomplete and would distort monthly trends.

The period also gives a like-for-like comparison of January to June 2025 against January to June 2026.

1.4 Loading and cleaning

The data was loaded into MySQL 8 with two setup scripts, kept separate from the 75 analysis queries. The raw tables are left untouched, so every cleaning step can be traced.

1. **Load raw data** (`lfb_setup_load.sql`). Both CSV files were loaded with `LOAD DATA LOCAL INFILE` into `raw_incidents` and `raw_mobilisations`, every column as text. This stops rows being rejected because of blank cells. Column names were converted to snake_case, for example `Notional Cost (£)` became `notional_cost_gbp`.
2. **Build typed tables** (`lfb_setup_clean.sql`). Clean `incidents` and `mobilisations` tables were built from the raw tables:
    - Dates and times converted to real date and time types.
    - Durations, counts and cost converted to numbers.
    - Blank text converted to NULL.
3. **Limit to the period.** Incidents were filtered to 1 January 2025 to 30 June 2026.
4. **Link the tables.** Mobilisations were kept only where they match an incident in the clean table, using an inner join. This removes orphan mobilisations and aligns the period in one step.
5. **Remove duplicates.** Where a mobilisation ID appeared more than once, one row was kept.
6. **Add keys.** `incident_number` is the primary key of `incidents`, and `resource_mobilisation_id` the primary key of `mobilisations`. An index on `mobilisations.incident_number` supports the join.

Incidents with no linked mobilisation were kept, because they are real incidents (see 1.6). Queries that need mobilisation data use an inner join; queries about all incidents do not.

1.5 Column dictionary

Types are as stored in the clean tables. Role groups each column as an identifier, categorical, numeric measure, date/time or location. Meanings should be checked against the LFB metadata file published on each dataset page.

incidents (39 columns)

Column	Type	Role	Meaning
incident_number	VARCHAR	Identifier	Unique incident reference; primary key
date_of_call	DATE	Date/time	Date the call was received
cal_year	INT	Date/time	Calendar year of the call
time_of_call	TIME	Date/time	Time the call was received

Column	Type	Role	Meaning
hour_of_call	INT	Date/time	Hour of the call, 0 to 23
incident_group	VARCHAR	Categorical	Fire, Special Service or False Alarm
stop_code_description	VARCHAR	Categorical	More detailed incident category
special_service_type	VARCHAR	Categorical	Type of non-fire incident; filled for special services only
property_category	VARCHAR	Categorical	Broad property class, e.g. dwelling, road vehicle, outdoor
property_type	VARCHAR	Categorical	Detailed property type
address_qualifier	VARCHAR	Categorical	Where the incident was in relation to the address
postcode_full	VARCHAR	Location	Full postcode; blank for many incidents
postcode_district	VARCHAR	Location	First part of the postcode, e.g. SE1
uprn	VARCHAR	Identifier	Unique Property Reference Number
usrn	VARCHAR	Identifier	Unique Street Reference Number
borough_code	VARCHAR	Location	Borough code (code, not a readable name)
borough_name	VARCHAR	Location	Borough name, upper case
proper_case	VARCHAR	Location	Borough name, normal case
ward_code	VARCHAR	Location	Ward code (code, not a readable name)
ward_name	VARCHAR	Location	Ward name
ward_name_new	VARCHAR	Location	Ward name on

Column	Type	Role	Meaning
			updated boundaries
easting_m	DECIMAL	Location	Exact grid easting; blank for many incidents
northing_m	DECIMAL	Location	Exact grid northing; blank for many incidents
easting_rounded	INT	Location	Rounded grid easting
northing_rounded	INT	Location	Rounded grid northing
latitude	DECIMAL	Location	Latitude
longitude	DECIMAL	Location	Longitude
frs	VARCHAR	Categorical	Fire and rescue service area
incident_station_ground	VARCHAR	Categorical	Fire station area the incident fell in
first_pump_attendance_time	INT	Measure	Seconds until the first fire engine arrived
first_pump_deployed_from_station	VARCHAR	Categorical	Station the first engine came from
second_pump_attendance_time	INT	Measure	Seconds until the second fire engine arrived
second_pump_deployed_from_station	VARCHAR	Categorical	Station the second engine came from
num_stations_with_pumps_attending	INT	Measure	Number of stations that sent engines
num_pumps_attending	INT	Measure	Number of fire engines that attended
pump_count	INT	Measure	Total fire engine attendances
pump_minutes_rounded	INT	Measure	Total fire engine time at the incident, in minutes (rounded)

Column	Type	Role	Meaning
notional_cost_gbp	DECIMAL	Measure	Estimated cost of the attendance, in £
num_calls	INT	Measure	Number of calls received about the incident

mobilisations (24 columns)

Column	Type	Role	Meaning
incident_number	VARCHAR	Identifier	Links to incidents
cal_year	INT	Date/time	Calendar year of the call
borough_name	VARCHAR	Location	Borough of the incident
ward_name	VARCHAR	Location	Ward of the incident
hour_of_call	INT	Date/time	Hour of the call, 0 to 23
resource_mobilisation_id	VARCHAR	Identifier	Unique mobilisation reference; primary key
resource_code	VARCHAR	Identifier	Fire engine that was sent (call sign)
performance_reporting	VARCHAR	Categorical	Whether the mobilisation counts as first, second or later for performance reporting
date_time_mobilised	DATETIME	Date/time	When the engine was sent
date_time_mobile	DATETIME	Date/time	When the engine set off
date_time_arrived	DATETIME	Date/time	When the engine arrived
turnout_time_seconds	INT	Measure	Seconds from being sent to setting off
travel_time_seconds	INT	Measure	Seconds from setting off to arriving
attendance_time_seco	INT	Measure	Seconds from being

Column	Type	Role	Meaning
nds			sent to arriving
date_time_left	DATETIME	Date/time	When the engine left the incident
date_time_returned	DATETIME	Date/time	When the engine returned
deployed_from_station_code	VARCHAR	Categorical	Station code (code, not a readable name)
deployed_from_station_name	VARCHAR	Categorical	Station the engine was sent from
deployed_from_location	VARCHAR	Categorical	Whether the engine was at its home station or elsewhere
pump_order	INT	Measure	Order in which the engine arrived
plus_code_code	VARCHAR	Categorical	Code for initial or additional attendance
plus_code_description	VARCHAR	Categorical	Initial attendance or additional resources
delay_code_id	INT	Categorical	Delay reason code (code, not a readable value)
delay_code_description	VARCHAR	Categorical	Reason for any delay

1.6 Data quality issues

Twelve issues were found while loading, profiling and querying the data. None prevents the analysis, but several affect how queries must be written.

Issue	Scale	How it was handled
Files cover different periods	Incidents from Jan 2024, mobilisations from Jan 2025	Both limited to Jan 2025 to Jun 2026
Different date formats between files	Incidents `15-Jan-25`; mobilisations `15/01/2025 08:32`	Each converted with its own format pattern

Issue	Scale	How it was handled
Blank values	Common in second-pump, delay and location columns	Stored as NULL; averages ignore them
Mobilisations with no matching incident	1,415 rows	Excluded
Incidents with no linked mobilisation	12,460 incidents	Kept; 12,043 of them have no first pump time, so no fire engine was recorded
Incidents with a pump time but no mobilisation	417 incidents	Kept; a gap between the two files
Incidents with no incident group	125 incidents	Kept; shown as NULL
Duplicate mobilisation IDs	11 IDs appeared twice, with small differences (e.g. ward Thames View vs Barking Riverside)	One row kept per ID
Station areas remapped	Ten stations closed on 9 January 2014	LFB has applied post-2014 station areas to all records; no action needed
Return time almost never recorded	date_time_returned filled for 1 of 312,817 mobilisations; blank in the source file	Kept as NULL; Q25 and Q44 cannot be answered from this data (see their interpretations)
Additional-resource flag carries no information	plus_code_description holds a single value for all 312,817 mobilisations	Kept; escalation cannot be identified, so Q72 returns no escalations
Attendance times appear capped	No attendance time exceeds 1,200 seconds (20 minutes)	Kept; the slowest responses may be under-stated (Q31, Q33)

Open question: the orphan and no-mobilisation counts above were measured on January 2025 to July 2026 data. Recount them on the clean tables so they match the final period.

2. End Consumer

The end consumer is a London Fire Brigade resource planning lead. They decide which boroughs and stations need more fire engines or improvement, and where demand on the service can be reduced.

3. Analytical Perspectives

The five perspectives each answer a different question the planning lead needs settled before deciding where resources go. All five cover every incident type: fires, special services and false alarms.

#	Perspective	Rationale
1	Where incidents happen and how concentrated they are	Shows which boroughs, wards and station areas carry the most demand, so resources can be placed where they are most needed.
2	When incidents happen and how demand is changing	Shows the busiest months, days and hours, and whether demand is rising, so staff and fire engines on duty match peak demand.
3	How fast engines respond and what delays them	Measures how quickly engines arrive and what slows them down, so the causes of slow responses can be targeted and improved.
4	What incidents and call-outs cost	Shows the cost of incidents in pump time and money, by station area and incident type, so budgets can be set and spent efficiently.
5	What types of incidents occur and their impact	Breaks incidents down by type, including false alarms, and shows what each type demands in resources, so prevention can target the most costly ones.

4. Question Bank

The 75 questions were generated with AI using the Section 9 prompting sequence (Steps A to E), then checked against the real column list and the required concept list. Questions are numbered Q1 to Q75: Perspective 1 is Q1–Q15, Perspective 2 Q16–Q30, Perspective 3 Q31–Q45, Perspective 4 Q46–Q60 and Perspective 5 Q61–Q75.

4.1 Concept map

Every required concept appears across the 75, spread over all five perspectives. The concepts below are planned; each query's comment header will state what its final SQL actually uses.

# in perspective	Band	P1 Where	P2 When	P3 Response	P4 Cost	P5 Types

# in perspective	Band	P1 Where	P2 When	P3 Response	P4 Cost	P5 Types
1	Easy	SELECT	WHERE	WHERE	SELECT	SELECT
2	Easy	SELECT	WHERE	SELECT	WHERE	WHERE
3	Easy	WHERE	WHERE	WHERE	WHERE	WHERE
4	Easy	WHERE	WHERE	WHERE	WHERE	WHERE
5	Easy	WHERE	WHERE	WHERE	WHERE	WHERE
6	Medium	GROUP BY	GROUP BY	GROUP BY	GROUP BY	GROUP BY
7	Medium	GROUP BY	GROUP BY	GROUP BY	GROUP BY	GROUP BY
8	Medium	GROUP BY	GROUP BY	GROUP BY	GROUP BY	GROUP BY
9	Medium	Join	GROUP BY	Join	GROUP BY	Join
10	Medium	Join (LEFT)	Join	GROUP BY	Join	GROUP BY
11	Difficult	Window	Window	Join + Window	Window	CTE + Window
12	Difficult	VIEW (create) + Window	VIEW (create) + Window	VIEW (reuse)	VIEW (reuse) + Window	CTE + Join
13	Difficult	VIEW (create)	CTE + Window	Window	VIEW (reuse)	Window
14	Difficult	CTE	CTE + Window	CTE + Window	CTE + Join	CTE + Window
15	Difficult	CTE	VIEW (reuse) + Window	VIEW (reuse) + Window	VIEW (reuse)	CTE

Three views are each created once and reused:

- **Borough summary:** created in Q12, reused in Q45 and Q58.
- **Station ground summary:** created in Q13, reused in Q42 and Q60.
- **Daily summary:** created in Q27, reused in Q30 and Q57.

Across the 75, joins appear in 9 questions, window functions in 16 and CTEs in 10.

4.2 Perspective 1: Where incidents happen and how concentrated they are

Q	Band	Question

Q	Band	Question
Q1	Easy	Which London boroughs appear in the incident records? List each borough once, in alphabetical order.
Q2	Easy	What were the 10 incidents that needed the most fire engines? For each, show the incident number, date of call, borough, ward, property type and number of pumps attending, largest first.
Q3	Easy	Which fire incidents happened in Westminster, Camden or Tower Hamlets between 1 January and 30 June 2026? Show the date, ward, property type and station ground, most recent first.
Q4	Easy	Which incidents have a postcode district recorded but no full postcode? Show the incident number, borough, postcode district and incident group, so we can see where location precision is weakest.
Q5	Easy	Which incidents that were not false alarms happened in wards whose name contains “Park”? Show the ward, borough, incident group and date of call.
Q6	Medium	For each property category, how many incidents occurred and in how many different boroughs? Show only categories with more than 1,000 incidents, busiest first.
Q7	Medium	Which station grounds dealt with more than 2,000 incidents in the period? For each, show

Q	Band	Question
		the number of incidents and the number of distinct wards covered.
Q8	Medium	For each borough, how many incidents were fires, special services and false alarms? Show all three counts side by side on one row per borough.
Q9	Medium	Which boroughs received the most fire engine deployments? For each borough, show the total engines sent, and the average number of engines per incident, counting only incidents that had at least one engine mobilised.
Q10	Medium	In each borough, how many incidents have no fire engine mobilisation linked to them at all, and what proportion of that borough's incidents does that represent?
Q11	Difficult	Within each borough, which are the three wards with the most incidents? Rank the wards inside their own borough and show only the top three per borough.
Q12	Difficult	Create a reusable borough summary showing, for each borough, the number of incidents, the number with a recorded cost, the total notional cost and the average first pump attendance time. Using it, what share of all London incidents does each borough account for? Ordering boroughs from busiest to quietest, what is the running cumulative share, and how

Q	Band	Question
		many boroughs together make up half of all incidents?
Q13	Difficult	Create a reusable station ground summary showing, for each station ground, total incidents, fires, false alarms, the false alarm percentage, average first pump attendance time, average second pump attendance time (where a second engine attended) and total notional cost. Using it, which station grounds have a false alarm percentage above London's overall false alarm percentage, meaning all false alarms divided by all incidents?
Q14	Difficult	Which individual properties, identified by property reference number and ignoring incidents with no reference recorded, have been the site of five or more incidents? For each, show the borough, property type and incident count, along with the share of its borough's total incidents that this one property accounts for.
Q15	Difficult	Which wards have an incident count more than double the average ward in their own borough? Show the ward, borough, ward incident count and the borough's average incidents per ward, ordered by how far above average each ward is.

4.3 Perspective 2: When incidents happen and how demand is changing

Q	Band	Question
Q16	Easy	What incidents did the brigade handle on Bonfire Night, 5 November 2025? Show the time of call, incident group, detailed incident category and borough, in the order the calls came in.
Q17	Easy	Which incidents were reported on Christmas Day 2025, New Year's Eve 2025 or New Year's Day 2026? Show the date, time of call, incident group and borough, ordered by date and time.
Q18	Easy	Which incidents during July 2025 were called in the evening peak, between 17:00 and 19:59, and needed three or more pumps? Show the date, time of call, incident group, borough and number of pumps attending, most recent first.
Q19	Easy	Which false alarms at properties whose type mentions "school" were called during school hours, between 09:00 and 15:59? Show the date, time of call, property type, borough and detailed incident category, most recent first.
Q20	Easy	Which incidents in the last week of the data, 24 to 30 June 2026, used more than 60 pump minutes? Show the date, time of call, incident group, borough, number of pumps attending and pump minutes, highest pump minutes first.
Q21	Medium	How many incidents does the

Q	Band	Question
		brigade receive in each hour of the day, and what share of each hour's incidents are false alarms? Show only the hours where false alarms make up more than a third of incidents, busiest hour first.
Q22	Medium	How does demand vary by day of the week? For each weekday, show the total incidents and the number of fires, special services and false alarms, from busiest day to quietest.
Q23	Medium	Comparing January–June 2025 with January–June 2026, how many incidents of each incident group occurred in each period, and what is the change in number and in percentage?
Q24	Medium	Which station grounds have the highest share of their incidents called in overnight, between midnight and 05:59? Consider only station grounds with at least 500 incidents, and show total incidents, overnight incidents and the overnight share, highest share first.
Q25	Medium	For each incident group, how long on average is a fire engine committed to a call in each hour of the day, measured from mobilisation to returning? Exclude engines with no recorded return time, show the result in minutes, and order by incident group and hour.
Q26	Difficult	How did the monthly number of incidents change from one month to the next across all 18

Q	Band	Question
		months? Show each month's total, the previous month's total, and the change in number and in percentage.
Q27	Difficult	Create a reusable daily summary showing, for each date, the number of incidents, fires, special services and false alarms, and the total notional cost. Using it, which 10 periods saw the highest sustained demand based on a 7-day moving average of daily incidents? Count only complete 7-day windows, so start from 7 January 2025, and show the end date of each window, that day's count and the moving average.
Q28	Difficult	Which hours of the day saw the biggest growth in incidents between January–June 2025 and January–June 2026? For each hour of call, show the number of incidents in both periods and the change in number and percentage, then rank the hours from largest percentage increase to largest decrease.
Q29	Difficult	Which boroughs saw the biggest percentage increase in incidents between January–June 2025 and January–June 2026? Rank all boroughs by percentage change, from largest increase to largest decrease.
Q30	Difficult	Using the daily summary, which days had more than 50% more incidents than the average day in the same month of the same

Q	Band	Question
		year? For each of these surge days, show its count, that month's daily average and the following day's count, to see whether demand stayed high.

4.4 Perspective 3: How fast engines respond and what delays them

Q	Band	Question
Q31	Easy	Which incidents waited more than 20 minutes for the first fire engine to arrive? Show the incident number, date of call, borough, station ground and first pump attendance time in both seconds and minutes, slowest first.
Q32	Easy	What are the actual delay reasons recorded against fire engine mobilisations? List each reason once, excluding mobilisations with no delay reason recorded and those marked "Not held up", in alphabetical order.
Q33	Easy	Which mobilisations were held up by any traffic-related delay and still took longer than 8 minutes to arrive? Show the engine, the station it was sent from, the delay reason and the attendance time, slowest first.
Q34	Easy	Which incidents had the first engine arrive within the 6-minute target but the second engine arrive outside the 8-minute target? Show the incident number, borough, station ground, and first and second pump attendance times.

Q	Band	Question
Q35	Easy	What were the 25 longest turnout times for engines that were not at their home station when mobilised? Show the engine, the station it was deployed from, its deployment location and the turnout time in seconds.
Q36	Medium	How does the speed of the first engine vary by hour of the day? For each hour of call, show the number of incidents with a recorded first pump attendance time, the average attendance time in seconds, and the percentage arriving within 6 minutes, slowest hour first.
Q37	Medium	What does each actual delay reason cost in response time? For each reason, show the number of mobilisations affected and the average turnout, travel and attendance times, most frequent reason first.
Q38	Medium	Which stations are slowest to get their engines moving once mobilised? For each station engines are sent from, show the number of mobilisations and the average turnout and travel times. Show only stations with at least 1,000 mobilisations, slowest average turnout first.
Q39	Medium	In each borough, what share of engines counted as first for performance reporting had an actual delay recorded? Show the number of first engines, the number delayed and the

Q	Band	Question
		percentage delayed, highest percentage first.
Q40	Medium	Do engines respond faster when they are sent from their home station rather than from elsewhere? For each deployment location, show the number of mobilisations and the average turnout, travel and attendance times.
Q41	Difficult	For each actual delay reason, which three boroughs record the most mobilisations held up by that reason? Rank the boroughs within each delay reason and show only the top three per reason.
Q42	Difficult	Using the station ground summary, which station grounds miss both the 6-minute first-engine target and the 8-minute second-engine target on average? Show their incident counts and both average attendance times, worst first-engine average first.
Q43	Difficult	How has London's average first pump attendance time moved month by month across all 18 months? Show each month's average, the change from the previous month and a 3-month moving average. Show the moving average only from March 2025 onward, where three full months are available.
Q44	Difficult	For each fire engine, how much time typically passes between returning from one call and being mobilised to its next one?

Q	Band	Question
		Show each engine's number of mobilisations, its average gap in minutes and how often it was sent to a new call before it had returned from the previous one. Exclude mobilisations with no recorded return time, and list the busiest engines first.
Q45	Difficult	Using the borough summary, rank boroughs from slowest to fastest average first pump attendance time. For each borough, show its average, whether it misses the 6-minute target, the average of the next borough in the ranking and the gap in seconds between them, to see where the biggest step-changes in performance lie.

4.5 Perspective 4: What incidents and call-outs cost

Q	Band	Question
Q46	Easy	What were the 10 most expensive incidents in the period? Show the incident number, date of call, borough, incident group, property type, number of pumps attending, pump minutes and notional cost, most expensive first.
Q47	Easy	Which false alarms had a notional cost of more than £1,000? Show the date, borough, property type, detailed incident category and cost, highest cost first.
Q48	Easy	Which incidents have no notional cost recorded? Show the incident number, date of

Q	Band	Question
		call, incident group, borough and pump minutes, to see where cost data is missing.
Q49	Easy	Which special service incidents between January and June 2026 kept fire engines busy for between two and four hours in total pump time? Show the date, special service type, borough, pump minutes and notional cost, longest first.
Q50	Easy	Which fires in properties whose type mentions “flat” cost more than £5,000? Show the date, borough, ward, property type, number of pumps attending and notional cost, highest cost first.
Q51	Medium	How is the brigade’s notional cost split across incident groups? For each of fires, special services and false alarms, show the number of incidents, total cost, average cost per incident and total pump minutes.
Q52	Medium	Which 15 wards account for the most fire engine time? For each ward, show its borough, the number of incidents, total pump minutes, total notional cost and average pump minutes per incident, highest total pump minutes first.
Q53	Medium	How much do false alarms cost in each property category? Considering only false alarms, show the number of incidents, total cost and average cost per category. Include only categories where false alarms

Q	Band	Question
		cost more than £100,000 in total, highest first.
Q54	Medium	How does cost rise as more engines attend? For each number of pumps attending, show the number of incidents, the average notional cost and the average pump minutes, in order of pumps attending. Exclude incidents with no number of pumps recorded.
Q55	Medium	For each special service type, how many fire engines were sent and how long did they spend on scene on average, from arriving to leaving? Show the time in minutes, excluding engines with no recorded arrival or leaving time, longest average first.
Q56	Difficult	Within each borough, what were the five most expensive incidents? Rank incidents by notional cost inside their own borough and show the borough, rank, incident number, date, incident group, property type and cost.
Q57	Difficult	Using the daily summary, how did the brigade's total notional cost build up month by month across all 18 months? Show each month's total cost, the running cumulative total, and the change from the previous month in pounds and percentage.
Q58	Difficult	Using the borough summary, which boroughs have an average cost per incident above

Q	Band	Question
		London's overall average, meaning total cost divided by the number of incidents with a recorded cost? Show each borough's average, London's average and the difference, highest first.
Q59	Difficult	Do incidents where at least one engine had an actual delay cost more than incidents with no delay? Classify each incident that had at least one engine mobilised as delayed or not delayed, then compare the number of incidents, average notional cost and average pump minutes for the two groups.
Q60	Difficult	Using the station ground summary, which station grounds have both a false alarm percentage above London's overall false alarm percentage and a total notional cost above the average station ground? Show their false alarm percentage and total cost, highest cost first, as the places where reducing false alarms would free the most resource.

4.6 Perspective 5: What types of incidents occur and their impact

Q	Band	Question
Q61	Easy	What types of special service does the brigade attend? List each special service type once, excluding incidents with no type recorded, in alphabetical order.
Q62	Easy	Which fires prompted 10 or more calls from the public?

Q	Band	Question
		Show the date, time of call, borough, property type, number of calls and number of pumps attending, most calls first.
Q63	Easy	Which incidents needed engines from four or more different stations? Show the incident number, date, borough, incident group, property type, number of stations with pumps attending and number of pumps attending, largest first.
Q64	Easy	Which special service incidents had a type that mentions flooding? Show the date, borough, ward, property type, special service type and pump minutes, longest pump time first.
Q65	Easy	Which incidents were recorded as malicious false alarms between January and June 2026? Show the date, time of call, borough, ward and property type, most recent first.
Q66	Medium	What is the typical impact of each detailed incident category? For each category, show the number of incidents, the average number of pumps attending, the average pump minutes and the average number of calls received, highest average pump minutes first.
Q67	Medium	Which special service types most often need more than one fire engine? For each type with at least 500 incidents, show the

Q	Band	Question
		number of incidents and the percentage where two or more pumps attended, based on incidents with a recorded pump count, highest percentage first.
Q68	Medium	Which 10 property types have the most fires? For each, show the number of fires, the average number of pumps attending and the average pump minutes per fire.
Q69	Medium	Which types of incident most often draw fire engines from outside their home station? For each incident group, show the number of engines mobilised, the number not at their home station when mobilised, and that number as a percentage of all engines mobilised.
Q70	Medium	Where exactly do fires happen relative to the recorded address? Considering only fires, show for each address qualifier the number of fires, the average number of pumps attending and the average pump minutes, most common first.
Q71	Difficult	In each borough, what is the single most common type of special service? Show the borough, the special service type, its number of incidents and its share of that borough's special service incidents. If two types tie, show both.
Q72	Difficult	How often do incidents in each property category escalate beyond the initial attendance? Classify each incident that had

Q	Band	Question
		at least one engine mobilised by whether any engine was sent as additional resources. For each property category, show the number of incidents, the number that escalated and the escalation percentage, and rank the categories from most to least likely to escalate.
Q73	Difficult	For incidents where at least three engines arrived, how long after the first engine did the third one arrive? Number the engines within each incident by their recorded arrival time (not the recorded pump order), excluding engines with no arrival time. Then, for each incident group, show the number of such incidents and the average gap in minutes between the first and third arrivals.
Q74	Difficult	Which properties suffer repeated false alarms in quick succession? For each property (by property reference number, ignoring incidents with no reference), find the gap in days between consecutive false alarms. List the properties with at least three false alarms that followed the previous one within 30 days, with their borough, property type and number of quick repeats, most repeats first.
Q75	Difficult	Which special service types take up a bigger share of the brigade's special service engine time than their share of special

Q	Band	Question
		service incidents? For each type, show its share of special service incidents, its share of special service pump minutes and the ratio between the two, highest ratio first.

5. Query Results and Interpretation

Each query's actual output is shown below its question. Results of up to 35 rows are shown in full; longer results show the first 20 rows with the total row count.

5.1 Perspective 1: Where incidents happen and how concentrated they are

Q1

Question: Which London boroughs appear in the incident records? List each borough once, in alphabetical order.

Concept: Basic SELECT - DISTINCT, ORDER BY

borough_name
BARKING AND DAGENHAM
BARNET
BEXLEY
BRENT
BROMLEY
CAMDEN
CITY OF LONDON
CROYDON
EALING
ENFIELD
GREENWICH
HACKNEY
HAMMERSMITH AND FULHAM
HARINGEY

borough_name
HARROW
HAVERING
HILLINGDON
HOUNSLOW
ISLINGTON
KENSINGTON AND CHELSEA
KINGSTON UPON THAMES
LAMBETH
LEWISHAM
MERTON
NEWHAM
REDBRIDGE
RICHMOND UPON THAMES
SOUTHWARK
SUTTON
TOWER HAMLETS
WALTHAM FOREST
WANDSWORTH
WESTMINSTER

33 rows returned (shown in full).

Interpretation: All 33 London boroughs (32 boroughs plus the City of London) appear, so the data covers the whole of London.

Q2

Question: What were the 10 incidents that needed the most fire engines? For each, show the incident number, date of call, borough, ward, property type and number of pumps attending, largest first.

Concept: Basic SELECT - ORDER BY, LIMIT

incident_number	date_of_call	borough_name	ward_name	property_type	num_pumps_attending

incident_number	date_of_call	borough_name	ward_name	property_type	num_pumps_attending
032078-23022025	2025-02-23	HOUNSLOW	CHISWICK HOMEFIELDS	Barge	10
066143-14042026	2026-04-14	HACKNEY	DE BEAUVOIR	Purpose Built Flats/Maisonettes - 10 or more storeys	10
036598-27022026	2026-02-27	HACKNEY	HAGGERSTON	Purpose Built Flats/Maisonettes - 10 or more storeys	9
006729-11012025	2025-01-11	BARKING AND DAGENHAM	BARKING RIVERSIDE	Outdoor storage	9
017262-27012026	2026-01-27	ISLINGTON	ST. PETER'S & CANALSIDE	Purpose Built Flats/Maisonettes - 4 to 9 storeys	8
056433-30032026	2026-03-30	TOWER HAMLETS	MILE END	Purpose Built Flats/Maisonettes - 4 to 9 storeys	8
049082-23032025	2025-03-23	WANDSWORTH	FALCONBROOK	Purpose Built Flats/Maisonettes - 4 to 9 storeys	8
034173-27022025	2025-02-27	LAMBETH	STREATHAM ST. LEONARD'S	Converted Flat/Maisonettes - 3 or more storeys	8
063274-09042026	2026-04-09	KENSINGTON AND CHELSEA	ROYAL HOSPITAL	Barge	8
076878-29042026	2026-04-29	BARKING AND DAGENHAM	ABBEY	Purpose Built Flats/Maisonettes - 10 or more storeys	8

10 rows returned (shown in full).

Interpretation: The largest incidents are mostly in flats and maisonettes, especially blocks of 10 or more storeys, and two barge fires needed 8–10 pumps. High-rise flats should be a priority for resource planning.

Q3

Question: Which fire incidents happened in Westminster, Camden or Tower Hamlets between 1 January and 30 June 2026? Show the date, ward, property type and station ground, most recent first.

Concept: Basic WHERE - IN, BETWEEN

date_of_call	borough_name	ward_name	property_type	incident_station_ground
2026-06-30	WESTMINSTER	WEST END	Small refuse/rubbish container	Soho
2026-06-30	CAMDEN	HIGHGATE	Purpose Built Flats/Maisonettes - 4 to 9 storeys	Kentish Town
2026-06-30	TOWER HAMLETS	CANARY WHARF	Domestic garden (vegetation not equipment)	Millwall
2026-06-30	TOWER HAMLETS	BETHNAL GREEN	Small refuse/rubbish container	Bethnal Green
2026-06-30	TOWER HAMLETS	BLACKWALL & CUBITT TOWN	Car	Millwall
2026-06-29	CAMDEN	ST. PANCRAS & SOMERS TOWN	Purpose built office	Euston
2026-06-29	CAMDEN	HAVERSTOCK	Converted Flat/Maisonettes - 3 or more storeys	Kentish Town
2026-06-29	WESTMINSTER	ST. JAMES'S	Small refuse/rubbish container	Soho
2026-06-29	CAMDEN	BLOOMSBURY	Telephone box	Euston
2026-06-29	WESTMINSTER	MAIDA VALE	Refuse/rubbish tip	Paddington
2026-06-29	CAMDEN	HAMPSTEAD	Van	West Hampstead

date_of_call	borough_name	ward_name	property_type	incident_station_ground
		TOWN		
2026-06-29	WESTMINSTER	PIMLICO SOUTH	Car	Lambeth
2026-06-28	TOWER HAMLETS	SHADWELL	Loose refuse	Shadwell
2026-06-28	WESTMINSTER	VINCENT SQUARE	Purpose Built Flats/Maisonettes - 4 to 9 storeys	Lambeth
2026-06-28	TOWER HAMLETS	BROMLEY SOUTH	Purpose Built Flats/Maisonettes - 4 to 9 storeys	Poplar
2026-06-27	WESTMINSTER	LANCASTER GATE	Small refuse/rubbish container	Paddington
2026-06-27	TOWER HAMLETS	SPITALFIELDS & BANGLATOWN	Hedge	Whitechapel
2026-06-27	TOWER HAMLETS	POPLAR	Road surface/pavement	Poplar
2026-06-27	WESTMINSTER	BAYSWATER	Small refuse/rubbish container	Paddington
2026-06-27	WESTMINSTER	WEST END	Restaurant/cafe	Soho

Showing 20 of [total] rows.

Interpretation: Fires in these three central boroughs are frequent, and the most recent records are dominated by small refuse and rubbish container fires, with the West End and the Soho station ground appearing often.

Q4

Question: Which incidents have a postcode district recorded but no full postcode? Show the incident number, borough, postcode district and incident group, so we can see where location precision is weakest.

Concept: Basic WHERE - IS NULL / IS NOT NULL

incident_number	borough_name	postcode_district	incident_group
000001-01012026	LAMBETH	SW16	False Alarm

incident_number	borough_name	postcode_district	incident_group
000004-01012025	HAMMERSMITH AND FULHAM	SW6	False Alarm
000005-01012025	MERTON	KT3	Fire
000008-01012026	ISLINGTON	N7	Fire
000009-01012026	KENSINGTON AND CHELSEA	W11	False Alarm
000013-01012026	WANDSWORTH	SW18	Special Service
000022-01012026	HARINGEY	N22	Fire
000026-01012025	HOUNSLOW	W4	Special Service
000028-01012025	BRENT	HA3	Fire
000036-01012026	HARINGEY	N22	False Alarm
000037-01012025	BROMLEY	SE19	Fire
000039-01012026	CROYDON	CR0	Special Service
000040-01012025	LAMBETH	SW9	False Alarm
000044-01012025	NEWHAM	E16	Fire
000049-01012026	EALING	UB5	Fire
000051-01012026	NEWHAM	E16	False Alarm
000052-01012026	LEWISHAM	SE6	False Alarm
000054-01012026	LAMBETH	SW9	False Alarm
000055-01012026	CAMDEN	WC1X	Special Service
000064-01012026	HARINGEY	N15	Fire

Showing 20 of [total] rows.

Interpretation: Many incidents have only a postcode district and no full postcode, across all boroughs and incident groups, so location recording needs to improve for precise planning.

Q5

Question: Which incidents that were not false alarms happened in wards whose name contains "Park"? Show the ward, borough, incident group and date of call.

Concept: Basic WHERE - LIKE, NOT

ward_name	borough_name	incident_group	date_of_call
QUEENS PARK	BRENT	Fire	2025-01-01
TUFNELL PARK	ISLINGTON	Fire	2026-01-01
WEMBLEY PARK	BRENT	Special Service	2026-01-01
REGENT'S PARK	CAMDEN	Fire	2026-01-01
HATCH LANE & HIGHAMS PARK NORTH	WALTHAM FOREST	Fire	2026-01-01
WANSTEAD PARK	REDBRIDGE	Fire	2025-01-01
FINSBURY PARK	ISLINGTON	Fire	2025-01-01
REGENT'S PARK	WESTMINSTER	Fire	2025-01-01
NORTHUMBERLAND PARK	HARINGEY	Fire	2026-01-01
FINSBURY PARK	ISLINGTON	Fire	2026-01-01
GREENWICH PARK	GREENWICH	Fire	2026-01-01
HALE END & HIGHAMS PARK SOUTH	WALTHAM FOREST	Fire	2025-01-01
FINSBURY PARK	ISLINGTON	Fire	2025-01-01
HYDE PARK	WESTMINSTER	Special Service	2025-01-01
BATTERSEA PARK	WANDSWORTH	Special Service	2026-01-01
NORTHWICK PARK	BRENT	Special Service	2026-01-01
FINSBURY PARK	ISLINGTON	Fire	2026-01-01
BUSH HILL PARK	ENFIELD	Special Service	2026-01-01
RAYNES PARK	MERTON	Fire	2025-01-01
BRONDESBURY PARK	BRENT	Special Service	2026-01-01

Showing 20 of [total] rows.

Interpretation: Incidents in “Park” wards are a mix of fires and special services, spread across many wards and boroughs, so these wards share no single pattern of demand.

Q6

Question: For each property category, how many incidents occurred and in how many different boroughs? Show only categories with more than 1,000 incidents, busiest first.

Concept: GROUP BY / aggregation with HAVING

property_category	incidents	boroughs
Dwelling	131658	33
Non Residential	29646	33
Outdoor	16442	33
Road Vehicle	11610	33
Other Residential	10330	33
Outdoor Structure	9703	33

6 rows returned (shown in full).

Interpretation: Dwellings are by far the most common location (131,658 incidents, about 63%) and occur in every borough, so home fire safety should be the top prevention priority.

Q7

Question: Which station grounds dealt with more than 2,000 incidents in the period? For each, show the number of incidents and the number of distinct wards covered.

Concept: GROUP BY / aggregation with HAVING

incident_station_ground	incidents	wards_covered
Paddington	5106	14
Soho	4916	9
Lambeth	4181	14
Euston	4135	13
North Kensington	3947	18
West Hampstead	3664	18
Tottenham	3557	16
Shoreditch	3428	20
Chelsea	3396	12
Stoke Newington	3323	16
Hammersmith	3311	20
Plumstead	3108	14
Croydon	3067	13

incident_station_ground	incidents	wards_covered
Kentish Town	3045	16
Ilford	3014	20
Holloway	2943	17
Homerton	2925	12
Bethnal Green	2898	14
Brixton	2869	13
West Norwood	2779	16

Showing 20 of 47 rows.

Interpretation: 47 station grounds dealt with more than 2,000 incidents, led by Paddington (5,106) and Soho (4,916), showing central London is the main pressure point.

Q8

Question: For each borough, how many incidents were fires, special services and false alarms? Show all three counts side by side on one row per borough.

Concept: GROUP BY / conditional aggregation

borough_name	fires	special_services	false_alarms
BARKING AND DAGENHAM	935	1908	1647
BARNET	1103	3057	3603
BEXLEY	841	1781	1551
BRENT	1163	3001	2791
BROMLEY	1118	2478	2626
CAMDEN	905	3408	4685
CITY OF LONDON	128	344	1003
CROYDON	1125	3983	3561
EALING	1135	2988	3036
ENFIELD	1090	2840	2025
GREENWICH	1144	3023	2582
HACKNEY	920	4073	3327

borough_name	fires	special_services	false_alarms
HAMMERSMITH AND FULHAM	552	2131	2945
HARINGEY	962	2662	2483
HARROW	610	1601	1630
HAVERING	1025	1744	2112
HILLINGDON	1070	2393	3104
HOUNSLOW	945	2002	2336
ISLINGTON	746	3307	3088
KENSINGTON AND CHELSEA	427	2539	4095
KINGSTON UPON THAMES	411	1021	1231
LAMBETH	1039	4148	4183
LEWISHAM	889	3531	3343
MERTON	507	1378	1544
NEWHAM	1334	3052	2721
REDBRIDGE	909	2066	1909
RICHMOND UPON THAMES	430	1290	1410
SOUTHWARK	1058	4270	3922
SUTTON	464	1304	2200
TOWER HAMLETS	1047	4362	3668
WALTHAM FOREST	993	2104	2621
WANDSWORTH	747	2804	3189
WESTMINSTER	1411	4258	7565

33 rows returned (shown in full).

Interpretation: False alarms are the most common call-out overall, especially in central boroughs such as Westminster (7,565), Camden (4,685) and Kensington and Chelsea (4,095), followed closely by special services.

Q9

Question: Which boroughs received the most fire engine deployments? For each borough, show the total engines sent, and the average number of engines per incident, counting only incidents that had at least one engine mobilised.

Concept: Join (INNER JOIN) with aggregation

borough_name	total_engines_sent	avg_engines
WESTMINSTER	20913	1.69
TOWER HAMLETS	14762	1.79
LAMBETH	14212	1.62
SOUTHWARK	14174	1.66
CAMDEN	13876	1.65
CROYDON	12932	1.55
LEWISHAM	12032	1.62
HACKNEY	12024	1.58
BARNET	11273	1.51
KENSINGTON AND CHELSEA	11145	1.68
WANDSWORTH	11062	1.72
BRENT	10549	1.60
NEWHAM	10535	1.57
ISLINGTON	10413	1.57
EALING	10225	1.50
GREENWICH	10127	1.58
HILLINGDON	9756	1.55
HAMMERSMITH AND FULHAM	9018	1.66
HARINGEY	8832	1.50
BROMLEY	8425	1.40
WALTHAM FOREST	8183	1.48
ENFIELD	8165	1.45

borough_name	total_engines_sent	avg_engines
HOUNSLOW	7916	1.58
REDBRIDGE	7000	1.50
BARKING AND DAGENHAM	6810	1.61
HAVERING	6556	1.40
SUTTON	5755	1.50
BEXLEY	5596	1.40
HARROW	5526	1.50
MERTON	4695	1.42
RICHMOND UPON THAMES	4292	1.44
KINGSTON UPON THAMES	3665	1.45
CITY OF LONDON	2373	1.73

33 rows returned (shown in full).

Interpretation: Westminster received far more engine deployments than any other borough (20,913). Boroughs average 1.4 to 1.8 engines per incident, so most incidents need one or two engines.

Q10

Question: In each borough, how many incidents have no fire engine mobilisation linked to them at all, and what proportion of that borough's incidents does that represent?

Concept: Join (LEFT JOIN) with aggregation

borough_name	total_incidents	no_mobilisation	pct_no_mobilisation
TOWER HAMLETS	9085	815	8.97
HACKNEY	8324	706	8.48
SOUTHWARK	9255	714	7.71
CITY OF LONDON	1476	106	7.18
ISLINGTON	7145	501	7.01
CAMDEN	9009	615	6.83
WESTMINSTER	13242	852	6.43
LAMBETH	9375	596	6.36

borough_name	total_incidents	no_mobilisation	pct_no_mobilisation
KENSINGTON AND CHELSEA	7068	423	5.98
BARKING AND DAGENHAM	4492	261	5.81
NEWHAM	7115	401	5.64
KINGSTON UPON THAMES	2664	144	5.41
BRENT	6958	373	5.36
ENFIELD	5957	309	5.19
HOUNSLOW	5290	272	5.14
GREENWICH	6753	326	4.83
WANDSWORTH	6740	322	4.78
RICHMOND UPON THAMES	3135	149	4.75
EALING	7163	338	4.72
REDBRIDGE	4886	219	4.48
LEWISHAM	7767	333	4.29
BEXLEY	4175	170	4.07
HAVERING	4883	197	4.03
HILLINGDON	6571	265	4.03
HARROW	3845	155	4.03
CROYDON	8675	347	4.00
HAMMERSMITH AND FULHAM	5631	209	3.71
BROMLEY	6226	228	3.66
HARINGEY	6112	224	3.66
MERTON	3431	125	3.64
BARNET	7766	282	3.63
WALTHAM FOREST	5723	201	3.51

borough_name	total_incidents	no_mobilisation	pct_no_mobilisation
SUTTON	3971	123	3.10

33 rows returned (shown in full).

Interpretation: Between 3.1% and 9.0% of each borough's incidents had no engine mobilisation recorded, highest in Tower Hamlets and Hackney.

Q11

Question: Within each borough, which are the three wards with the most incidents? Rank the wards inside their own borough and show only the top three per borough.

Concept: Window function - ranking within a partition

borough_name	ward_rank	ward_name	incidents
BARKING AND DAGENHAM	1	GASCOIGNE	409
BARKING AND DAGENHAM	2	VILLAGE	330
BARKING AND DAGENHAM	3	NORTHBURY	307
BARNET	1	HENDON	529
BARNET	2	CHILDS HILL	508
BARNET	3	BURNT OAK	477
BEXLEY	1	ERITH	457
BEXLEY	2	SIDCUP	390
BEXLEY	3	THAMESMEAD EAST	375
BRENT	1	KILBURN	709
BRENT	2	STONEBRIDGE	607
BRENT	3	DOLLIS HILL	510
BROMLEY	1	BROMLEY TOWN	500
BROMLEY	2	ST. MARY CRAY	484
BROMLEY	3	PENGE & CATOR	436
CAMDEN	1	BLOOMSBURY	1219
CAMDEN	2	HOLBORN & COVENT	964

borough_name	ward_rank	ward_name	incidents
		GARDEN	
CAMDEN	3	REGENT'S PARK	704
CITY OF LONDON	1	BISHOPSGATE	153
CITY OF LONDON	2	FARRINGDON WITHIN	121

Showing 20 of 100 rows.

Interpretation: This shows the wards to focus on within each borough. West End (2,596) and St. James's (2,024) are far above every other ward. Tied wards share a rank, so 100 rows are returned rather than 99.

Q12

Question: Create a reusable borough summary showing, for each borough, the number of incidents, the number with a recorded cost, the total notional cost and the average first pump attendance time. Using it, what share of all London incidents does each borough account for? Ordering boroughs from busiest to quietest, what is the running cumulative share, and how many boroughs together make up half of all incidents?

Concept: VIEW (creates borough summary) + window function (running total)

borough_rank	borough_name	incidents	share_pct	cumulative_share_pct
1	WESTMINSTER	13242	6.31	6.31
2	LAMBETH	9375	4.47	10.77
3	SOUTHWARK	9255	4.41	15.18
4	TOWER HAMLETS	9085	4.33	19.51
5	CAMDEN	9009	4.29	23.80
6	CROYDON	8675	4.13	27.94
7	HACKNEY	8324	3.97	31.90
8	LEWISHAM	7767	3.70	35.60
9	BARNET	7766	3.70	39.30
10	EALING	7163	3.41	42.71
11	ISLINGTON	7145	3.40	46.12
12	NEWHAM	7115	3.39	49.51

borough_rank	borough_name	incidents	share_pct	cumulative_share_pct
13	KENSINGTON AND CHELSEA	7068	3.37	52.88
14	BRENT	6958	3.31	56.19
15	GREENWICH	6753	3.22	59.41
16	WANDSWORTH	6740	3.21	62.62
17	HILLINGDON	6571	3.13	65.75
18	BROMLEY	6226	2.97	68.71
19	HARINGEY	6112	2.91	71.63
20	ENFIELD	5957	2.84	74.46
21	WALTHAM FOREST	5723	2.73	77.19
22	HAMMERSMITH AND FULHAM	5631	2.68	79.87
23	HOUNSLOW	5290	2.52	82.39
24	REDBRIDGE	4886	2.33	84.72
25	HAVERING	4883	2.33	87.05
26	BARKING AND DAGENHAM	4492	2.14	89.19
27	BEXLEY	4175	1.99	91.18
28	SUTTON	3971	1.89	93.07
29	HARROW	3845	1.83	94.90
30	MERTON	3431	1.63	96.53
31	RICHMOND UPON THAMES	3135	1.49	98.03
32	KINGSTON UPON THAMES	2664	1.27	99.30
33	CITY OF LONDON	1476	0.70	100.00

33 rows returned (shown in full).

Interpretation: Westminster alone accounts for 6.3% of all incidents, and 13 of the 33 boroughs together make up half of London's incidents, so demand is concentrated in a minority of areas.

Q13

Question: Create a reusable station ground summary showing, for each station ground, total incidents, fires, false alarms, the false alarm percentage, average first pump attendance time, average second pump attendance time (where a second engine attended) and total notional cost. Using it, which station grounds have a false alarm percentage above London's overall false alarm percentage, meaning all false alarms divided by all incidents?

Concept: VIEW (creates station ground summary) + subquery

incident_station_ground	total_incidents	false_alarms	false_alarm_pct
Soho	4916	2999	61.00
Kensington	2169	1306	60.21
Chelsea	3396	2044	60.19
Dowgate	1861	1089	58.52
Wallington	1501	859	57.23
Euston	4135	2324	56.20
Wimbledon	1384	772	55.78
Whitechapel	1465	783	53.45
Sutton	2123	1133	53.37
Fulham	2097	1118	53.31
Heathrow	998	531	53.21
Richmond	1229	652	53.05
Paddington	5106	2708	53.04
Hammersmith	3311	1756	53.04
Barnet	1497	780	52.10
Ruislip	1424	734	51.54
North Kensington	3947	2028	51.38
Tooting	2441	1239	50.76
Leytonstone	1490	740	49.66

incident_station_ground	total_incidents	false_alarms	false_alarm_pct
Romford	1265	623	49.25

Showing 20 of 39 rows.

Interpretation: 39 station grounds have a false alarm rate above the London average. Soho, Kensington and Chelsea are highest at around 60%, so false alarm reduction should start with central stations, although some outer grounds such as Wallington and Sutton are also high.

Q14

Question: Which individual properties, identified by property reference number and ignoring incidents with no reference recorded, have been the site of five or more incidents? For each, show the borough, property type and incident count, along with the share of its borough's total incidents that this one property accounts for.

Concept: CTE

uprn	borough_name	property_type	incidents	pct_of_borough
10070703694	HAVERING	Woodland/forest - broadleaf/hardwood	155	3.174
5124020	CAMDEN	Road surface/pavement	133	1.476
10059101565	HILLINGDON	Roadside vegetation	90	1.370
10010227695	GREENWICH	Prison	78	1.155
46255350	NEWHAM	Hospital	76	1.068
10059101564	HILLINGDON	Van	70	1.065
100023398319	HOUNSLOW	Young offenders unit	54	1.021
5087826	CAMDEN	Underground car park	54	0.599
10059101572	HILLINGDON	Passenger plane	50	0.761
10033555561	WESTMINSTER	Single shop	48	0.362
10059101545	HILLINGDON	Small refuse/rubbish	46	0.700

uprn	borough_name	property_type	incidents	pct_of_borough
		container		
100023249499	GREENWICH	Secondary school	45	0.666
6741884	TOWER HAMLETS	Hotel/motel	45	0.495
5155897	CAMDEN	Hospital	44	0.488
5133289	CAMDEN	Medical/health centre	43	0.477
121050999	WANDSWORTH	Prison	42	0.623
10093393950	LEWISHAM	Purpose built office	41	0.528
5199558	CAMDEN	Student Hall of Residence	41	0.455
10059101553	HILLINGDON	Small refuse/rubbish container	40	0.609
100023401310	RICHMOND UPON THAMES	Road surface/pavement	40	1.276

Showing 20 of 1,674 rows.

Interpretation: Some properties have repeated incidents, the worst being a Havering woodland (155) and a Camden road (133). Prisons, hospitals and Heathrow-area sites also recur, so these properties need targeted prevention.

Q15

Question: Which wards have an incident count more than double the average ward in their own borough? Show the ward, borough, ward incident count and the borough's average incidents per ward, ordered by how far above average each ward is.

Concept: CTE

ward_name	borough_name	incidents	borough_avg_per_ward	times_above_avg
HEATHROW VILLAGES	HILLINGDON	1246	273.8	4.55
WEST END	WESTMINSTER	2596	662.1	3.92
KINGSTON TOWN	KINGSTON UPON THAMES	496	140.2	3.54

ward_name	borough_name	incidents	borough_avg_per_ward	times_above_avg
BLOOMSBURY	CAMDEN	1219	391.7	3.11
ST. JAMES'S	WESTMINSTER	2024	662.1	3.06
FAIRFIELD	CROYDON	887	299.1	2.97
BISHOPSGATE	CITY OF LONDON	153	56.8	2.70
KILBURN	BRENT	709	267.6	2.65
SOUTH RICHMOND	RICHMOND UPON THAMES	384	149.3	2.57
SUTTON CENTRAL	SUTTON	485	189.1	2.56
HOLBORN & COVENT GARDEN	CAMDEN	964	391.7	2.46
GREENHILL	HARROW	430	174.8	2.46
ILFORD TOWN	REDBRIDGE	459	187.9	2.44
SOUTH ACTON	EALING	645	265.3	2.43
CRICKET GREEN	MERTON	378	156.0	2.42
RAINHAM & WENNINGTON	HAVERING	563	232.5	2.42
STONEBRIDGE	BRENT	607	267.6	2.27
STRATFORD	NEWHAM	597	263.5	2.27
WADDON	CROYDON	669	299.1	2.24
EDMONTON GREEN	ENFIELD	510	229.1	2.23
UXBRIDGE	HILLINGDON	606	273.8	2.21
DEPTFORD	LEWISHAM	738	337.7	2.19
FARRINGDON WITHIN	CITY OF LONDON	121	56.8	2.13
CASTLE BAYNARD	CITY OF LONDON	120	56.8	2.11
COLEMAN STREET	CITY OF LONDON	120	56.8	2.11
OLD KENT ROAD	SOUTHWARK	721	342.8	2.10

ward_name	borough_name	incidents	borough_avg_per_ward	times_above_avg
WHITECHAPEL	TOWER HAMLETS	855	413.0	2.07
LONDON BRIDGE & WEST BERMONDSEY	SOUTHWARK	688	342.8	2.01

28 rows returned (shown in full).

Interpretation: 28 wards have more than double their borough's average, led by Heathrow Villages (4.5 times) and West End (3.9 times), mostly town centres and transport hubs.

5.2 Perspective 2: When incidents happen and how demand is changing

Q16

Question: What incidents did the brigade handle on Bonfire Night, 5 November 2025? Show the time of call, incident group, detailed incident category and borough, in the order the calls came in.

Concept: Basic WHERE - date filter, ORDER BY

time_of_call	incident_group	stop_code_description	borough_name
00:01:57	False Alarm	AFA	HILLINGDON
00:02:37	False Alarm	AFA	HOUNSLOW
00:13:10	False Alarm	False alarm - Good intent	LAMBETH
00:17:03	Fire	Primary Fire	BROMLEY
00:21:05	False Alarm	AFA	HILLINGDON
00:40:10	Special Service	Special Service	SOUTHWARK
00:54:25	Special Service	Special Service	SOUTHWARK
01:05:59	Fire	Secondary Fire	EALING
01:07:16	Fire	Secondary Fire	TOWER HAMLETS
01:12:43	Special Service	Special Service	BRENT
01:20:05	Special Service	Special Service	ISLINGTON
01:31:09	Special Service	Special Service	BROMLEY
01:33:14	Fire	Secondary Fire	HARROW
01:43:52	False Alarm	AFA	WANDSWORTH

time_of_call	incident_group	stop_code_description	borough_name
02:07:10	False Alarm	AFA	HAMMERSMITH AND FULHAM
02:08:19	False Alarm	AFA	BROMLEY
02:12:17	False Alarm	AFA	NEWHAM
02:15:59	False Alarm	False alarm - Good intent	GREENWICH
02:18:31	Special Service	Special Service	HACKNEY
02:21:32	False Alarm	AFA	NEWHAM

Showing 20 of 475 rows.

Interpretation: Bonfire Night (475 incidents) is steady through the day, mostly false alarms and special services. From late afternoon, secondary fires rise sharply across many boroughs, peaking between 20:00 and 23:00, so evening cover is key.

Q17

Question: Which incidents were reported on Christmas Day 2025, New Year's Eve 2025 or New Year's Day 2026? Show the date, time of call, incident group and borough, ordered by date and time.

Concept: Basic WHERE - IN

date_of_call	time_of_call	incident_group	borough_name
2025-12-25	00:03:04	False Alarm	HILLINGDON
2025-12-25	00:10:54	False Alarm	HACKNEY
2025-12-25	00:11:23	False Alarm	TOWER HAMLETS
2025-12-25	00:16:57	Fire	BROMLEY
2025-12-25	00:21:05	False Alarm	HAMMERSMITH AND FULHAM
2025-12-25	00:23:41	Special Service	BRENT
2025-12-25	00:24:56	Special Service	BARKING AND DAGENHAM
2025-12-25	00:25:00	False Alarm	HAMMERSMITH AND FULHAM
2025-12-25	00:25:34	Fire	BRENT
2025-12-25	00:25:43	False Alarm	HARROW

date_of_call	time_of_call	incident_group	borough_name
2025-12-25	00:26:07	Fire	WESTMINSTER
2025-12-25	00:31:26	Special Service	CAMDEN
2025-12-25	00:32:22	Special Service	BROMLEY
2025-12-25	00:33:32	False Alarm	BROMLEY
2025-12-25	00:35:03	False Alarm	REDBRIDGE
2025-12-25	00:45:31	False Alarm	HARROW
2025-12-25	00:47:00	False Alarm	BRENT
2025-12-25	00:53:17	False Alarm	WALTHAM FOREST
2025-12-25	00:57:31	False Alarm	HACKNEY
2025-12-25	01:02:30	Special Service	REDBRIDGE

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: Christmas Day calls are mostly special services and false alarms, spread across many boroughs, so full cover is still needed over the holiday period.

Q18

Question: Which incidents during July 2025 were called in the evening peak, between 17:00 and 19:59, and needed three or more pumps? Show the date, time of call, incident group, borough and number of pumps attending, most recent first.

Concept: Basic WHERE - BETWEEN, AND

date_of_call	time_of_call	incident_group	borough_name	num_pumps_attending
2025-07-31	19:52:51	False Alarm	REDBRIDGE	3
2025-07-31	19:23:31	False Alarm	LAMBETH	3
2025-07-31	19:09:02	False Alarm	REDBRIDGE	5
2025-07-31	18:19:12	False Alarm	MERTON	4
2025-07-31	17:06:20	False Alarm	SOUTHWARK	4
2025-07-30	17:51:15	Fire	SUTTON	5
2025-07-30	17:26:01	False Alarm	TOWER HAMLETS	5
2025-07-29	19:54:52	Fire	KINGSTON UPON THAMES	3

date_of_call	time_of_call	incident_group	borough_name	num_pumps_attending
2025-07-29	19:47:53	False Alarm	CAMDEN	5
2025-07-29	19:21:12	False Alarm	WANDSWORTH	3
2025-07-29	17:29:17	False Alarm	CROYDON	3
2025-07-29	17:13:50	False Alarm	HILLINGDON	5
2025-07-28	19:48:38	False Alarm	KENSINGTON AND CHELSEA	3
2025-07-28	19:28:16	False Alarm	KENSINGTON AND CHELSEA	3
2025-07-28	19:20:57	False Alarm	EALING	3
2025-07-28	18:35:32	False Alarm	CROYDON	5
2025-07-28	18:25:22	Fire	HAVERING	3
2025-07-28	17:50:13	False Alarm	WESTMINSTER	5
2025-07-28	17:47:21	False Alarm	WESTMINSTER	5
2025-07-27	19:48:32	False Alarm	HACKNEY	5

Showing 20 of 164 rows.

Interpretation: Most July evening-peak incidents needing three or more pumps were false alarms, often with five pumps attending, so a lot of resource is used on calls that turn out not to be real.

Q19

Question: Which false alarms at properties whose type mentions "school" were called during school hours, between 09:00 and 15:59? Show the date, time of call, property type, borough and detailed incident category, most recent first.

Concept: Basic WHERE - LIKE, BETWEEN

date_of_call	time_of_call	property_type	borough_name	stop_code_description
2026-06-30	13:49:17	Secondary school	REDBRIDGE	AFA
2026-06-30	13:23:29	Secondary school	CAMDEN	AFA
2026-06-30	11:26:59	Infant/Primary school	KENSINGTON AND CHELSEA	AFA
2026-06-30	09:33:18	Infant/Primary	HILLINGDON	AFA

date_of_call	time_of_call	property_type	borough_name	stop_code_description
		school		
2026-06-30	09:30:57	Secondary school	WALTHAM FOREST	AFA
2026-06-29	14:11:33	Infant/Primary school	BARNET	AFA
2026-06-29	13:59:28	Secondary school	SUTTON	AFA
2026-06-27	15:39:53	Infant/Primary school	EALING	AFA
2026-06-27	14:30:47	Infant/Primary school	BROMLEY	AFA
2026-06-27	12:00:50	Secondary school	LAMBETH	AFA
2026-06-26	14:33:06	Infant/Primary school	HARROW	AFA
2026-06-26	13:41:23	Infant/Primary school	HARINGEY	AFA
2026-06-26	13:05:00	Infant/Primary school	GREENWICH	AFA
2026-06-26	12:41:02	Pre School/nursery	CAMDEN	AFA
2026-06-26	11:22:17	Infant/Primary school	HACKNEY	AFA
2026-06-25	13:07:55	Pre School/nursery	KENSINGTON AND CHELSEA	False alarm - Good intent
2026-06-25	12:03:46	Secondary school	WANDSWORTH	AFA
2026-06-25	10:57:29	Infant/Primary school	KENSINGTON AND CHELSEA	AFA
2026-06-25	10:02:11	Infant/Primary school	SUTTON	AFA
2026-06-24	13:38:03	Secondary school	SOUTHWARK	AFA

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: School false alarms during school hours are almost all automatic fire alarms (AFAs), mostly in primary schools, so schools' alarm systems should be reviewed.

Q20

Question: Which incidents in the last week of the data, 24 to 30 June 2026, used more than 60 pump minutes? Show the date, time of call, incident group, borough, number of pumps attending and pump minutes, highest pump minutes first.

Concept: Basic WHERE - BETWEEN, comparison

date_of_call	time_of_call	incident_group	borough_name	num_pumps_attending	pump_minutes_rounded
2026-06-25	20:54:44	Fire	SOUTHWARK	4	7889
2026-06-24	18:23:39	Fire	WESTMINSTER	4	4383
2026-06-28	16:51:29	Fire	HACKNEY	2	3288
2026-06-26	13:24:20	Fire	WALTHAM FOREST	2	1778
2026-06-24	13:21:01	Fire	BRENT	4	1597
2026-06-28	18:42:33	Fire	HOUNSLOW	6	1283
2026-06-27	04:17:41	Fire	WANDSWORTH	2	905
2026-06-28	16:18:57	Fire	BEXLEY	6	890
2026-06-26	06:38:29	Fire	NEWHAM	2	881
2026-06-26	16:05:06	Fire	HOUNSLOW	2	845
2026-06-26	23:34:50	Fire	HAVERING	2	839
2026-06-29	20:22:50	Fire	KINGSTON UPON THAMES	2	794
2026-06-29	18:26:54	Fire	HAVERING	5	675
2026-06-29	08:19:08	Fire	WANDSWORTH	2	662
2026-06-25	08:56:17	Fire	WANDSWORTH	7	624
2026-06-29	08:54:27	Fire	HAMMERSMITH AND	4	611

date_of_call	time_of_call	incident_group	borough_name	num_pumps_attending	pump_minutes_rounded
			FULHAM		
2026-06-24	05:52:32	Fire	SUTTON	4	573
2026-06-28	10:29:29	Fire	EALING	2	564
2026-06-26	11:54:03	False Alarm	WALTHAM FOREST	5	516
2026-06-28	21:02:30	Fire	SOUTHWARK	2	468

Showing 20 of 457 rows.

Interpretation: In the last week of June, a few fires used huge amounts of pump time, notably Southwark (7,889 minutes) and Westminster (4,383 minutes), showing that a small number of large fires take up much of the engine time.

Q21

Question: How many incidents does the brigade receive in each hour of the day, and what share of each hour's incidents are false alarms? Show only the hours where false alarms make up more than a third of incidents, busiest hour first.

Concept: GROUP BY / aggregation with HAVING

hour_of_call	incidents	false_alarms	false_alarm_pct
18	12562	5353	42.61
17	12487	5016	40.17
19	12357	5143	41.62
20	12213	5492	44.97
16	11901	4654	39.11
15	11398	4552	39.94
21	11364	5152	45.34
13	11056	4949	44.76
14	11025	4902	44.46
12	10584	4838	45.71
11	10149	4881	48.09
22	9989	4439	44.44

hour_of_call	incidents	false_alarms	false_alarm_pct
10	9270	4555	49.14
23	8551	3838	44.88
9	8234	4089	49.66
0	7257	3174	43.74
8	6685	3088	46.19
1	6217	2700	43.43
7	5190	2360	45.47
2	4992	2323	46.53
3	4341	2062	47.50
6	4253	2353	55.33
4	4137	1940	46.89
5	3696	1883	50.95

24 rows returned (shown in full).

Interpretation: False alarms make up more than a third of incidents in every hour of the day. 17:00–20:00 is the busiest period, and 06:00 has the highest false alarm share (55%).

Q22

Question: How does demand vary by day of the week? For each weekday, show the total incidents and the number of fires, special services and false alarms, from busiest day to quietest.

Concept: GROUP BY / conditional aggregation

weekday	total_incidents	fires	special_services	false_alarms
Saturday	31048	4201	13551	13272
Friday	30731	4205	12684	13824
Tuesday	30114	4149	12206	13736
Wednesday	30038	4242	11997	13780
Thursday	29475	4065	12045	13346
Sunday	29415	4125	12560	12713
Monday	29087	4196	11808	13065

7 rows returned (shown in full).

Interpretation: Demand is fairly even across the week. Saturday is busiest (31,048) and has the most special services, and Monday is quietest (29,087).

Q23

Question: Comparing January-June 2025 with January-June 2026, how many incidents of each incident group occurred in each period, and what is the change in number and in percentage?

Concept: GROUP BY / conditional aggregation

incident_group	incidents_2025	incidents_2026	change_num	change_pct
NULL	0	117	117	NULL
False Alarm	29120	31985	2865	9.84
Fire	10104	9539	-565	-5.59
Special Service	27395	30286	2891	10.55

4 rows returned (shown in full).

Interpretation: From January–June 2025 to January–June 2026, special services rose 10.6% and false alarms 9.8%, while fires fell 5.6%, so demand is shifting away from fires. 117 incidents in 2026 have no incident group recorded.

Q24

Question: Which station grounds have the highest share of their incidents called in overnight, between midnight and 05:59? Consider only station grounds with at least 500 incidents, and show total incidents, overnight incidents and the overnight share, highest share first.

Concept: GROUP BY / aggregation with HAVING

incident_station_ground	total_incidents	overnight_incidents	overnight_pct
Dowgate	1861	335	18.00
Soho	4916	867	17.64
Poplar	2762	473	17.13
Lewisham	2072	354	17.08
Dagenham	2740	467	17.04
Ilford	3014	512	16.99
Tottenham	3557	600	16.87
Holloway	2943	494	16.79

incident_station_ground	total_incidents	overnight_incidents	overnight_pct
Plaistow	2099	349	16.63
Whitechapel	1465	243	16.59
Old Kent Road	2227	368	16.52
Brixton	2869	467	16.28
Paddington	5106	827	16.20
Dockhead	1388	221	15.92
Stratford	2587	411	15.89
Wennington	963	153	15.89
Norbury	2563	407	15.88
Kensington	2169	339	15.63
Deptford	1531	239	15.61
Hendon	1682	262	15.58

Showing 20 of 101 rows.

Interpretation: Between 10% and 18% of each station ground's incidents happen overnight, highest at central stations such as Dowgate and Soho and lowest at outer stations such as Addington and Orpington.

Q25

Question: For each incident group, how long on average is a fire engine committed to a call in each hour of the day, measured from mobilisation to returning? Exclude engines with no recorded return time, show the result in minutes, and order by incident group and hour.

Concept: Join (INNER JOIN) with aggregation

incident_group	hour_of_call	engines	avg_minutes_committed
False Alarm	20	1	21.0

1 row returned (shown in full).

Interpretation: Only one row is returned because the source data records a return time for just 1 of 312,817 mobilisations (see section 1.6). The question cannot be answered from this dataset.

Q26

Question: How did the monthly number of incidents change from one month to the next across all 18 months? Show each month's total, the previous month's total, and the change in number and in percentage.

Concept: Window function - LAG

month	incidents	prev_month	change_num	change_pct
2025-01	11292	NULL	NULL	NULL
2025-02	9320	11292	-1972	-17.46
2025-03	10978	9320	1658	17.79
2025-04	11107	10978	129	1.18
2025-05	11827	11107	720	6.48
2025-06	12095	11827	268	2.27
2025-07	12699	12095	604	4.99
2025-08	12104	12699	-595	-4.69
2025-09	11379	12104	-725	-5.99
2025-10	11883	11379	504	4.43
2025-11	11662	11883	-221	-1.86
2025-12	11635	11662	-27	-0.23
2026-01	12105	11635	470	4.04
2026-02	10006	12105	-2099	-17.34
2026-03	11450	10006	1444	14.43
2026-04	11821	11450	371	3.24
2026-05	13528	11821	1707	14.44
2026-06	13017	13528	-511	-3.78

18 rows returned (shown in full).

Interpretation: Incidents fall by about 17% each February then recover, and May 2026 was the busiest month (13,528), showing a clear seasonal pattern and growth in 2026.

Q27

Question: Create a reusable daily summary showing, for each date, the number of incidents, fires, special services and false alarms, and the total notional cost. Using it, which 10 periods saw the highest

sustained demand based on a 7-day moving average of daily incidents? Count only complete 7-day windows, so start from 7 January 2025, and show the end date of each window, that day's count and the moving average.

Concept: VIEW (creates daily summary) + window function (moving average)

window_end_date	incidents	moving_avg_7
2026-06-29	482	552.4
2026-06-28	485	549.6
2026-06-27	505	540.4
2026-05-28	498	534.0
2026-06-26	585	531.7
2026-05-29	480	531.0
2026-05-30	488	530.7
2026-05-27	520	521.9
2026-05-31	463	517.6
2026-06-30	456	513.7

10 rows returned (shown in full).

Interpretation: All ten periods of highest sustained demand fall in late May and late June 2026, with the 7-day average peaking at 552 incidents a day, so early summer is the highest-pressure period.

Q28

Question: Which hours of the day saw the biggest growth in incidents between January-June 2025 and January-June 2026? For each hour of call, show the number of incidents in both periods and the change in number and percentage, then rank the hours from largest percentage increase to largest decrease.

Concept: CTE + window function (RANK)

growth_rank	hour_of_call	incidents_2025	incidents_2026	change_num	change_pct
1	4	1283	1457	174	13.56
2	7	1589	1804	215	13.53
3	8	2062	2299	237	11.49
4	13	3482	3871	389	11.17
5	22	3131	3466	335	10.70

growth_rank	hour_of_call	incidents_2025	incidents_2026	change_num	change_pct
6	5	1130	1237	107	9.47
7	3	1362	1487	125	9.18
8	14	3475	3792	317	9.12
9	19	3874	4226	352	9.09
10	15	3642	3941	299	8.21
11	9	2602	2812	210	8.07
12	0	2330	2509	179	7.68
13	18	4031	4334	303	7.52
14	10	2918	3133	215	7.37
14	11	3241	3480	239	7.37
16	21	3606	3865	259	7.18
17	2	1615	1730	115	7.12
18	20	3875	4146	271	6.99
19	16	3798	4055	257	6.77
20	17	4050	4302	252	6.22
21	6	1358	1440	82	6.04
22	23	2723	2884	161	5.91
23	12	3405	3567	162	4.76
24	1	2037	2090	53	2.60

24 rows returned (shown in full).

Interpretation: Every hour of the day saw growth. Early morning hours rose most (04:00 and 07:00, about 13.5%) and 01:00 least (2.6%).

Q29

Question: Which boroughs saw the biggest percentage increase in incidents between January-June 2025 and January-June 2026? Rank all boroughs by percentage change, from largest increase to largest decrease.

Concept: CTE + window function (RANK)

growth_rank	borough_name	incidents_2025	incidents_2026	change_num	change_pct
1	KINGSTON UPON THAMES	794	939	145	18.26
2	BARNET	2366	2781	415	17.54
3	BRENT	2149	2438	289	13.45
4	HACKNEY	2571	2903	332	12.91
5	MERTON	1054	1190	136	12.90
6	SOUTHWARK	2918	3260	342	11.72
7	REDBRIDGE	1493	1653	160	10.72
8	ISLINGTON	2278	2512	234	10.27
9	WESTMINSTER	4157	4582	425	10.22
10	HAVERING	1502	1647	145	9.65
11	WANDSWORTH	2119	2316	197	9.30
12	HARROW	1199	1309	110	9.17
13	RICHMOND UPON THAMES	1008	1094	86	8.53
14	CROYDON	2761	2996	235	8.51
15	TOWER HAMLETS	2884	3128	244	8.46
16	WALTHAM FOREST	1854	2004	150	8.09
17	LAMBETH	2961	3188	227	7.67
17	NEWHAM	2256	2429	173	7.67
19	BEXLEY	1344	1446	102	7.59
20	KENSINGTON AND CHELSEA	2272	2440	168	7.39
21	GREENWICH	2138	2284	146	6.83

growth_rank	borough_name	incidents_2025	incidents_2026	change_num	change_pct
22	CAMDEN	2831	3018	187	6.61
23	BROMLEY	1990	2115	125	6.28
24	CITY OF LONDON	483	513	30	6.21
25	EALING	2337	2464	127	5.43
26	HARINGEY	1982	2078	96	4.84
26	HOUNSLOW	1672	1753	81	4.84
28	SUTTON	1277	1318	41	3.21
29	ENFIELD	1939	1998	59	3.04
30	BARKING AND DAGENHAM	1436	1479	43	2.99
30	HILLINGDON	2144	2208	64	2.99
32	LEWISHAM	2564	2620	56	2.18
33	HAMMERSMITH AND FULHAM	1886	1824	-62	-3.29

33 rows returned (shown in full).

Interpretation: Every borough grew except Hammersmith and Fulham (−3.3%). Kingston upon Thames (+18.3%) and Barnet (+17.5%) grew most, so some outer boroughs may need more resources.

Q30

Question: Using the daily summary, which days had more than 50% more incidents than the average day in the same month of the same year? For each of these surge days, show its count, that month's daily average and the following day's count, to see whether demand stayed high.

Concept: VIEW (reuses daily summary) + window function (LEAD)

date_of_call	incidents	month_daily_avg	next_day_incidents
2026-06-23	727	433.9	581

1 row returned (shown in full).

Interpretation: Only one surge day was found: 23 June 2026, with 727 incidents against a monthly average of 434. Demand stayed high the next day (581), so surges can last more than one day.

5.3 Perspective 3: How fast engines respond and what delays them

Q31

Question: Which incidents waited more than 20 minutes for the first fire engine to arrive? Show the incident number, date of call, borough, station ground and first pump attendance time in both seconds and minutes, slowest first.

Concept: Basic WHERE - comparison, calculated column

incident_number	date_of_call	borough_name	incident_station_ground	attendance_seconds	attendance_minutes
078976-01052026	2026-05-01	ENFIELD	Enfield	1200	20.0
121673-04072025	2025-07-04	HOUNSLOW	Twickenham	1200	20.0

2 rows returned (shown in full).

Interpretation: Only two incidents show exactly 20 minutes (1,200 seconds), which suggests attendance times are capped at 1,200 seconds in the data, so the true slowest responses may be hidden.

Q32

Question: What are the actual delay reasons recorded against fire engine mobilisations? List each reason once, excluding mobilisations with no delay reason recorded and those marked "Not held up", in alphabetical order.

Concept: Basic SELECT - DISTINCT with WHERE

delay_code_description
Address incomplete/wrong
Appliance/Equipment defect
Arrived but held up - Other reason
At drills when mobilised
Mob/Radio problems when mobilised
On outside duty when mobilised
Traffic calming measures
Traffic, roadworks, etc
Weather conditions

9 rows returned (shown in full).

Interpretation: There are nine actual delay reasons, from traffic and roadworks to equipment defects and wrong addresses, so delays come from both outside factors and internal issues.

Q33

Question: Which mobilisations were held up by any traffic-related delay and still took longer than 8 minutes to arrive? Show the engine, the station it was sent from, the delay reason and the attendance time, slowest first.

Concept: Basic WHERE - LIKE, AND

resource_code	deployed_from_station_name	delay_code_description	attendance_time_seconds
E341	Dockhead	Traffic, roadworks, etc	1200
H412	Kingston	Traffic calming measures	1200
F361	Walthamstow	Traffic, roadworks, etc	1197
E361	Deptford	Traffic, roadworks, etc	1196
H221	Lambeth	Traffic, roadworks, etc	1194
G301	Wembley	Traffic, roadworks, etc	1194
A341	Edmonton	Traffic calming measures	1192
G272	North Kensington	Traffic, roadworks, etc	1192
G251	Ealing	Traffic, roadworks, etc	1190
G242	Southall	Traffic, roadworks, etc	1186
F301	Leytonstone	Traffic, roadworks, etc	1186
A342	Edmonton	Traffic, roadworks, etc	1185
A311	Holloway	Traffic, roadworks, etc	1183
A242	Soho	Traffic, roadworks, etc	1182
A431	Kentish Town	Traffic, roadworks, etc	1181
G312	Northolt	Traffic, roadworks, etc	1181
A331	Tottenham	Traffic, roadworks, etc	1180
E411	Orpington	Traffic calming measures	1180

resource_code	deployed_from_station_name	delay_code_description	attendance_time_seconds
G362	Hammersmith	Traffic calming measures	1179
F301	Leytonstone	Traffic, roadworks, etc	1179

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: Many engines held up by traffic still took well over 8 minutes, with many close to the 20-minute cap, so traffic is a major cause of slow responses across London.

Q34

Question: Which incidents had the first engine arrive within the 6-minute target but the second engine arrive outside the 8-minute target? Show the incident number, borough, station ground, and first and second pump attendance times.

Concept: Basic WHERE - AND conditions

incident_number	borough_name	incident_station_ground	first_pump_attendance_time	second_pump_attendance_time
000101-01012026	CAMDEN	Kentish Town	288	597
000153-01012025	BEXLEY	Erith	348	545
000200-01012025	BRENT	Wembley	25	847
000226-01012026	SOUTHWARK	Dockhead	251	553
000241-01012025	CAMDEN	Euston	271	565
000292-01012026	ISLINGTON	Holloway	245	486
000306-01012025	WANDSWORTH	Wandsworth	212	576
000367-01012026	BARNET	Barnet	172	656
000443-01012025	BARKING AND DAGENHAM	Dagenham	225	532
000455-01012026	HILLINGDON	Heathrow	250	554
000464-01012026	ISLINGTON	Holloway	237	556
000465-01012026	HAVERING	Harold Hill	279	504
000583-01012026	HACKNEY	Homerton	216	500
000603-01012026	WESTMINSTER	Paddington	165	481

incident_number	borough_name	incident_station_ground	first_pump_attendance_time	second_pump_attendance_time
000655-01012026	WESTMINSTER	Lambeth	225	633
000663-01012025	GREENWICH	Eltham	335	806
000664-01012025	BROMLEY	Biggin Hill	169	602
000744-01012026	SUTTON	Wallington	199	512
000768-01012025	MERTON	Wimbledon	356	498
000814-01012025	ENFIELD	Southgate	241	527

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: Many incidents had the first engine on time but the second late, some by several minutes, at both central and outer stations, so second-engine cover needs improving.

Q35

Question: What were the 25 longest turnout times for engines that were not at their home station when mobilised? Show the engine, the station it was deployed from, its deployment location and the turnout time in seconds.

Concept: Basic WHERE - ORDER BY, LIMIT

resource_code	deployed_from_station_name	deployed_from_location	turnout_time_seconds
E291	East Greenwich	Other Station	824
F282	Soho	Other Station	802
F362	Stratford	Other Station	783
G392	Heston	Other Station	744
E251	Eltham	Other Station	741
A242	North Kensington	Other Station	735
A301	Euston	Other Station	721
G362	North Kensington	Other Station	692
F331	Dowgate	Other Station	682
H322	West Norwood	Other Station	680
E361	Plumstead	Other Station	675
A322	Stoke Newington	Other Station	648

resource_code	deployed_from_station_name	deployed_from_location	turnout_time_seconds
E221	Deptford	Other Station	645
A241	Chelsea	Other Station	642
H312	Tooting	Other Station	631
E351	Poplar	Other Station	621
H381	Esher	Other Station	620
G222	Wembley	Other Station	619
G362	Kentish Town	Other Station	598
H222	Soho	Other Station	587
G361	North Kensington	Other Station	581
A242	Shoreditch	Other Station	561
H331	Wimbledon	Other Station	536
H291	Croydon	Other Station	536
H271	Wandsworth	Other Station	526

25 rows returned (shown in full).

Interpretation: Engines mobilised away from their home station can take almost 14 minutes to turn out (East Greenwich, 824 seconds), so moving engines between stations can cause long delays.

Q36

Question: How does the speed of the first engine vary by hour of the day? For each hour of call, show the number of incidents with a recorded first pump attendance time, the average attendance time in seconds, and the percentage arriving within 6 minutes, slowest hour first.

Concept: GROUP BY / conditional aggregation

hour_of_call	incidents	avg_attendance_seconds	pct_within_6_min
16	11136	348.1	60.62
17	11592	348.0	61.15
15	10651	347.9	60.02
13	10418	344.9	60.60
11	9620	344.2	61.31

hour_of_call	incidents	avg_attendance_seconds	pct_within_6_min
14	10384	343.7	61.56
12	9998	343.1	60.41
4	4003	342.8	64.28
18	11785	338.5	63.60
6	4074	338.4	64.65
5	3562	336.0	65.97
3	4181	335.0	65.44
2	4806	333.2	66.85
1	5983	328.1	68.33
7	4902	327.8	68.01
19	11601	325.9	67.39
10	8818	325.1	67.50
0	6974	320.2	70.68
8	6294	318.4	69.46
9	7796	311.7	71.82
23	8174	310.3	73.50
21	10781	308.5	73.56
20	11561	307.7	73.20
22	9512	305.8	74.47

24 rows returned (shown in full).

Interpretation: Afternoons are slowest for the first engine (about 348 seconds, only 60% within 6 minutes), while late evenings are fastest (about 306 seconds, 74% within 6 minutes), likely because of daytime traffic.

Q37

Question: What does each actual delay reason cost in response time? For each reason, show the number of mobilisations affected and the average turnout, travel and attendance times, most frequent reason first.

Concept: GROUP BY / aggregation

delay_code_description	mobilisations	avg_turnout	avg_travel	avg_attendance
Traffic, roadworks, etc	20622	78.0	436.1	514.1
Traffic calming measures	6743	83.7	413.8	497.6
Address incomplete/wrong	2986	85.5	490.4	575.8
On outside duty when mobilised	775	67.9	443.4	511.3
Arrived but held up - Other reason	727	92.9	465.3	558.1
Mob/Radio problems when mobilised	502	148.5	406.3	555.1
Weather conditions	408	90.2	403.1	493.3
Appliance/Equipment defect	369	101.6	470.0	571.7
At drills when mobilised	165	125.6	366.3	492.0

9 rows returned (shown in full).

Interpretation: Traffic and roadworks is by far the most common delay (20,622), but wrong or incomplete addresses add the most time (575.8 seconds average attendance), and radio problems almost double turnout time.

Q38

Question: Which stations are slowest to get their engines moving once mobilised? For each station engines are sent from, show the number of mobilisations and the average turnout and travel times. Show only stations with at least 1,000 mobilisations, slowest average turnout first.

Concept: GROUP BY / aggregation with HAVING

deployed_from_station_name	mobilisations	avg_turnout	avg_travel
Dagenham	4075	95.9	327.5

deployed_from_station_name	mobilisations	avg_turnout	avg_travel
Orpington	2002	91.1	315.5
Walthamstow	3540	90.2	283.8
Plumstead	4199	90.1	275.0
Greenwich	2559	89.5	281.5
Lee Green	2209	89.5	283.3
Edmonton	2931	89.2	312.5
Bromley	2952	87.0	312.6
Woodford	1558	85.9	328.8
Deptford	2246	85.2	274.5
Sidcup	2071	84.2	335.8
Eltham	1873	83.5	300.6
West Norwood	4094	82.8	295.0
Heston	3151	82.6	305.3
Beckenham	2427	82.6	308.2
Leytonstone	2495	82.5	317.4
Millwall	1875	82.0	286.0
New Cross	2858	81.6	276.6
Surbiton	1574	81.5	279.5
Plaistow	3190	81.5	296.5

Showing 20 of 100 rows.

Interpretation: Dagenham has the slowest turnout (95.9 seconds) and Acton and Peckham the fastest (63 seconds), while outer stations such as Heathrow and Hayes have the longest travel times.

Q39

Question: In each borough, what share of engines counted as first for performance reporting had an actual delay recorded? Show the number of first engines, the number delayed and the percentage delayed, highest percentage first.

Concept: Join (INNER JOIN) with aggregation

borough_name	first_engines	num_delayed	pct_delayed
BRENT	6585	1269	19.27
REDBRIDGE	4667	894	19.16
HAVERING	4686	875	18.67
EALING	6825	1192	17.47
HARINGEY	5888	981	16.66
RICHMOND UPON THAMES	2986	488	16.34
ENFIELD	5648	918	16.25
CAMDEN	8394	1328	15.82
BARNET	7484	1101	14.71
HARROW	3690	537	14.55
WESTMINSTER	12390	1772	14.30
KINGSTON UPON THAMES	2520	359	14.25
BARKING AND DAGENHAM	4231	600	14.18
HILLINGDON	6306	881	13.97
SUTTON	3848	537	13.96
NEWHAM	6714	921	13.72
WALTHAM FOREST	5522	748	13.55
HAMMERSMITH AND FULHAM	5422	660	12.17
HOUNSLOW	5018	601	11.98
ISLINGTON	6644	791	11.91
HACKNEY	7618	878	11.53
BROMLEY	5998	691	11.52
MERTON	3306	380	11.49
WANDSWORTH	6418	699	10.89

borough_name	first_engines	num_delayed	pct_delayed
CROYDON	8328	887	10.65
GREENWICH	6427	658	10.24
SOUTHWARK	8541	873	10.22
BEXLEY	4005	409	10.21
LEWISHAM	7434	721	9.70
TOWER HAMLETS	8270	791	9.56
CITY OF LONDON	1370	128	9.34
LAMBETH	8779	687	7.83
KENSINGTON AND CHELSEA	6645	519	7.81

33 rows returned (shown in full).

Interpretation: Between 7.8% and 19.3% of first engines were delayed, highest in Brent and Redbridge and lowest in Kensington and Chelsea and Lambeth.

Q40

Question: Do engines respond faster when they are sent from their home station rather than from elsewhere? For each deployment location, show the number of mobilisations and the average turnout, travel and attendance times.

Concept: GROUP BY / aggregation

deployed_from_location	mobilisations	avg_turnout	avg_travel	avg_attendance
Home Station	301701	77.4	295.2	371.5
Other Station	10601	48.4	332.9	377.6
NULL	515	97.8	285.8	367.1

3 rows returned (shown in full).

Interpretation: Almost all engines (96.4%) are sent from their home station. Engines sent from elsewhere turn out faster but travel further, so overall attendance is about the same (around 6 minutes).

Q41

Question: For each actual delay reason, which three boroughs record the most mobilisations held up by that reason? Rank the boroughs within each delay reason and show only the top three per reason.

Concept: Join + window function - ranking within a partition

delay_code_description	borough_rank	borough_name	mobilisations
Address incomplete/wrong	1	HILLINGDON	244
Address incomplete/wrong	2	HAVERING	206
Address incomplete/wrong	3	ENFIELD	184
Appliance/Equipment defect	1	BARNET	46
Appliance/Equipment defect	2	NEWHAM	21
Appliance/Equipment defect	3	WESTMINSTER	18
Arrived but held up - Other reason	1	HILLINGDON	56
Arrived but held up - Other reason	2	REDBRIDGE	55
Arrived but held up - Other reason	3	LEWISHAM	43
At drills when mobilised	1	CROYDON	17
At drills when mobilised	2	EALING	13
At drills when mobilised	3	HAMMERSMITH AND FULHAM	11
Mob/Radio problems when mobilised	1	SUTTON	57
Mob/Radio problems when mobilised	2	TOWER HAMLETS	26
Mob/Radio problems when mobilised	3	HACKNEY	25
On outside duty when	1	HILLINGDON	51

delay_code_description	borough_rank	borough_name	mobilisations
mobilised			
On outside duty when mobilised	2	GREENWICH	46
On outside duty when mobilised	2	BARNET	46
Traffic calming measures	1	HARINGEY	395
Traffic calming measures	2	ENFIELD	385
Traffic calming measures	3	BRENT	376
Traffic, roadworks, etc	1	WESTMINSTER	1671
Traffic, roadworks, etc	2	CAMDEN	1275
Traffic, roadworks, etc	3	BRENT	1097
Weather conditions	1	CROYDON	30
Weather conditions	2	EALING	27
Weather conditions	3	HILLINGDON	26

27 rows returned (shown in full).

Interpretation: Westminster has the most traffic delays (1,671), Hillingdon the most wrong-address delays (244), and Sutton stands out for radio problems (57), so each borough has different delay issues to fix.

Q42

Question: Using the station ground summary, which station grounds miss both the 6-minute first-engine target and the 8-minute second-engine target on average? Show their incident counts and both average attendance times, worst first-engine average first.

Concept: VIEW (reuses station ground summary)

incident_station_ground	total_incidents	avg_first_pump_attendance	avg_second_pump_attendance
Wennington	963	418.7	559.6
Biggin Hill	311	410.8	694.6

incident_station_ground	total_incidents	avg_first_pump_attendance	avg_second_pump_attendance
Ruislip	1424	404.8	529.1
Orpington	1612	395.2	485.0
Heathrow	998	391.1	542.8
Hayes	1640	384.7	489.4
Hornchurch	1482	383.3	538.4
Hainault	761	380.2	532.9
Woodford	1017	377.4	516.8
Southgate	1668	370.7	512.0
Harold Hill	982	366.5	509.4
Bexley	1394	362.9	498.6
Harrow	1815	362.4	484.5
Purley	1038	362.0	517.2

14 rows returned (shown in full).

Interpretation: 14 station grounds miss both the 6- and 8-minute targets, all in outer London, with Wennington slowest for the first engine (418.7 seconds) and Biggin Hill for the second (694.6 seconds).

Q43

Question: How has London's average first pump attendance time moved month by month across all 18 months? Show each month's average, the change from the previous month and a 3-month moving average. Show the moving average only from March 2025 onward, where three full months are available.

Concept: Window function - LAG and moving average

month	avg_first_pump_attendance	change_from_prev	moving_avg_3m
2025-01	317.0	NULL	NULL
2025-02	320.3	3.3	NULL
2025-03	322.7	2.5	320.0
2025-04	323.8	1.1	322.3
2025-05	328.7	4.8	325.1

month	avg_first_pump_attendance	change_from_prev	moving_avg_3m
2025-06	332.9	4.3	328.5
2025-07	343.4	10.4	335.0
2025-08	329.2	-14.2	335.2
2025-09	332.7	3.6	335.1
2025-10	332.9	0.1	331.6
2025-11	334.5	1.6	333.4
2025-12	325.9	-8.6	331.1
2026-01	328.2	2.4	329.5
2026-02	327.3	-1.0	327.1
2026-03	330.8	3.5	328.8
2026-04	332.7	1.9	330.3
2026-05	342.3	9.5	335.3
2026-06	339.4	-2.9	338.1

18 rows returned (shown in full).

Interpretation: Average first-engine attendance has risen from 317 seconds in January 2025 to about 339 in June 2026, with summer peaks in July 2025 and May 2026, so response times are getting worse.

Q44

Question: For each fire engine, how much time typically passes between returning from one call and being mobilised to its next one? Show each engine's number of mobilisations, its average gap in minutes and how often it was sent to a new call before it had returned from the previous one. Exclude mobilisations with no recorded return time, and list the busiest engines first.

Concept: CTE + window function - LAG within a partition

resource_code	mobilisations	avg_gap_minutes	sent_before_returning
F261	1	NULL	0

1 row returned (shown in full).

Interpretation: As with Q25, only one row is returned because return times are missing for all but one mobilisation in the source data (see section 1.6), so turnaround between calls cannot be measured from this dataset.

Q45

Question: Using the borough summary, rank boroughs from slowest to fastest average first pump attendance time. For each borough, show its average, whether it misses the 6-minute target, the average of the next borough in the ranking and the gap in seconds between them, to see where the biggest step-changes in performance lie.

Concept: VIEW (reuses borough summary) + window function (LEAD)

slowest_rank	borough_name	avg_first_pump_attendance	six_min_target	next_borough	next_borough_avg	gap_seconds
1	HILLINGDON	388.3	Misses target	BROMLEY	375.4	12.9
2	BROMLEY	375.4	Misses target	HAVERING	374.0	1.4
3	HAVERING	374.0	Misses target	ENFIELD	373.6	0.4
4	ENFIELD	373.6	Misses target	REDBRIDGE	368.3	5.3
5	REDBRIDGE	368.3	Misses target	RICHMOND UPON THAMES	362.6	5.7
6	RICHMOND UPON THAMES	362.6	Misses target	BARKING AND DAGENHAM	357.6	5.0
7	BARKING AND DAGENHAM	357.6	Meets target	BEXLEY	356.8	0.8
8	BEXLEY	356.8	Meets target	BRENT	355.9	0.9
9	BRENT	355.9	Meets target	HARROW	353.3	2.6
10	HARROW	353.3	Meets target	HOUNSLOW	351.4	1.9
11	HOUNSLOW	351.4	Meets target	EALING	350.0	1.4
12	EALING	350.0	Meets	KINGSTON	348.1	1.9

slowest_rank	borough_name	avg_first_pump_attendance	six_min_target	next_borough	next_borough_avg	gap_seconds
			target	UPON THAMES		
13	KINGSTON UPON THAMES	348.1	Meets target	BARNET	347.2	0.9
14	BARNET	347.2	Meets target	HARINGEY	335.9	11.3
15	HARINGEY	335.9	Meets target	MERTON	332.5	3.4
16	MERTON	332.5	Meets target	NEWHAM	331.6	0.9
17	NEWHAM	331.6	Meets target	GREENWICH	329.8	1.8
18	GREENWICH	329.8	Meets target	WALTHAM FOREST	328.8	1.0
19	WALTHAM FOREST	328.8	Meets target	SUTTON	324.8	4.0
20	SUTTON	324.8	Meets target	CROYDON	324.4	0.4
21	CROYDON	324.4	Meets target	WESTMINSTER	317.4	7.0
22	WESTMINSTER	317.4	Meets target	LEWISHAM	316.7	0.7
23	LEWISHAM	316.7	Meets target	WANDSWORTH	314.4	2.3
24	WANDSWORTH	314.4	Meets target	HACKNEY	312.6	1.8
25	HACKNEY	312.6	Meets target	CAMDEN	312.4	0.2
26	CAMDEN	312.4	Meets target	ISLINGTON	311.0	1.4

slowest_rank	borough_name	avg_first_pump_attendance	six_min_target	next_borough	next_borough_avg	gap_seconds
27	ISLINGTON	311.0	Meets target	SOUTHWARK	309.1	1.9
28	SOUTHWARK	309.1	Meets target	HAMMERSMITH AND FULHAM	307.4	1.7
29	HAMMERSMITH AND FULHAM	307.4	Meets target	TOWER HAMLETS	294.0	13.4
30	TOWER HAMLETS	294.0	Meets target	CITY OF LONDON	291.5	2.5
31	CITY OF LONDON	291.5	Meets target	LAMBETH	287.2	4.3
32	LAMBETH	287.2	Meets target	KENSINGTON AND CHELSEA	281.1	6.1
33	KENSINGTON AND CHELSEA	281.1	Meets target	NULL	NULL	NULL

33 rows returned (shown in full).

Interpretation: Six boroughs miss the 6-minute target, all outer boroughs, with Hillingdon slowest (388.3 seconds). The biggest jumps in performance are between Hammersmith and Fulham and Tower Hamlets, Hillingdon and Bromley, and Barnet and Haringey.

5.4 Perspective 4: What incidents and call-outs cost

Q46

Question: What were the 10 most expensive incidents in the period? Show the incident number, date of call, borough, incident group, property type, number of pumps attending, pump minutes and notional cost, most expensive first.

Concept: Basic SELECT - ORDER BY, LIMIT

incident_number	date_of_call	borough_name	incident_group	property_type	num_pumps_attending	pump_minutes_rounded	notional_cost_gbp
046906-	2025-03-	HILLINGD	Fire	Other	3	69016	494615.00

incident_number	date_of_call	borough_name	incident_group	property_type	num_pumps_attending	pump_minutes_rounded	notional_cost_gbp
20032025	20	ON		outdoor equipment/machinery			
128104-11072025	2025-07-11	BEXLEY	Fire	Recycling plant	3	57491	447472.00
221088-25112025	2025-11-25	EALING	Fire	Other retail	4	45193	351752.00
167849-06092025	2025-09-06	HAMMERSMITH AND FULHAM	Fire	Restaurant/cafe	5	32846	255651.00
109816-11062026	2026-06-11	BRENT	Fire	Hairdresser	4	29071	243712.00
098138-27052026	2026-05-27	BARNET	Fire	Large supermarket	2	26568	222728.00
079948-02052026	2026-05-02	ENFIELD	Fire	Warehouse	4	25019	209743.00
110780-21062025	2025-06-21	ENFIELD	Fire	Warehouse	2	26846	208951.00
156683-20082025	2025-08-20	LEWISHAM	Fire	Purpose Built Flats/Maisonettes - Up to 3 storeys	2	25903	201612.00
094414-29052025	2025-05-29	NEWHAM	Fire	Single shop	4	21466	167077.00

10 rows returned (shown in full).

Interpretation: The ten most expensive incidents are all fires, mostly at commercial and industrial sites, led by a Hillingdon fire costing £494,615, so a few large fires create very large costs.

Q47

Question: Which false alarms had a notional cost of more than £1,000? Show the date, borough, property type, detailed incident category and cost, highest cost first.

Concept: Basic WHERE - comparison

date_of_call	borough_name	property_type	stop_code_description	notional_cost_gbp
2025-09-22	HAVERING	Landfill site	False alarm - Good intent	14033.00
2025-04-30	KENSINGTON AND CHELSEA	Purpose Built Flats/Maisonettes - 4 to 9 storeys	AFA	9644.00
2026-05-14	CAMDEN	Secondary school	AFA	7335.00
2026-06-18	TOWER HAMLETS	Single shop	AFA	5047.00
2025-07-28	BRENT	Purpose Built Flats/Maisonettes - 4 to 9 storeys	False alarm - Good intent	4833.00
2025-09-09	WESTMINSTER	Converted Flat/Maisonettes - 3 or more storeys	False alarm - Good intent	4623.00
2026-06-03	BRENT	Purpose Built Flats/Maisonettes - 10 or more storeys	False alarm - Good intent	4619.00
2025-08-28	HAVERING	Scrub land	False alarm - Good intent	4491.00
2026-06-26	WALTHAM FOREST	Purpose Built Flats/Maisonettes - 4 to 9 storeys	AFA	4326.00
2026-06-22	HARROW	House - single occupancy	False alarm - Good intent	4007.00
2026-04-07	GREENWICH	Purpose Built Flats/Maisonettes - 4 to 9 storeys	AFA	3865.00
2026-01-08	HILLINGDON	Purpose Built Flats/Maisonettes	False alarm - Good intent	3674.00

date_of_call	borough_name	property_type	stop_code_description	notional_cost_gbp
		- 4 to 9 storeys		
2025-07-12	LAMBETH	Purpose Built Flats/Maisonettes - 4 to 9 storeys	AFA	3510.00
2025-10-01	BRENT	Self contained Sheltered Housing	AFA	3510.00
2026-06-08	LAMBETH	Purpose Built Flats/Maisonettes - 4 to 9 storeys	False alarm - Good intent	3487.00
2025-08-23	HILLINGDON	Hotel/motel	False alarm - Good intent	3471.00
2025-05-27	CITY OF LONDON	Hotel/motel	AFA	3394.00
2026-02-15	HILLINGDON	Self contained Sheltered Housing	AFA	3323.00
2026-05-13	BRENT	Takeaway/ fast food	False alarm - Good intent	3244.00
2026-02-21	NEWHAM	Purpose Built Flats/Maisonettes - 10 or more storeys	False alarm - Good intent	3191.00

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: Even false alarms can be expensive, up to £14,033 at a Havering landfill site, and many of the costliest are in purpose-built flats.

Q48

Question: Which incidents have no notional cost recorded? Show the incident number, date of call, incident group, borough and pump minutes, to see where cost data is missing.

Concept: Basic WHERE - IS NULL

incident_number	date_of_call	incident_group	borough_name	pump_minutes_rounded
(no rows)				

0 rows returned (no matching records).

Interpretation: No incidents are missing a notional cost, so cost data is complete for the whole period and cost totals can be trusted.

Q49

Question: Which special service incidents between January and June 2026 kept fire engines busy for between two and four hours in total pump time? Show the date, special service type, borough, pump minutes and notional cost, longest first.

Concept: Basic WHERE - BETWEEN

date_of_call	special_service_type	borough_name	pump_minutes_rounded	notional_cost_gbp
2026-01-13	Assist other agencies	HAMMERSMITH AND FULHAM	240	1868.00
2026-04-05	Assist other agencies	TOWER HAMLETS	240	2012.00
2026-06-27	Lift Release	WESTMINSTER	240	2012.00
2026-01-07	Flooding	WESTMINSTER	239	1860.00
2026-04-11	Spills and Leaks (not RTC)	WESTMINSTER	239	2004.00
2026-04-16	Flooding	HAVERING	239	2004.00
2026-05-06	Rescue or evacuation from water	TOWER HAMLETS	238	1995.00
2026-01-09	No action (not false alarm)	BARKING AND DAGENHAM	237	1845.00
2026-05-26	Hazardous Materials incident	LAMBETH	237	1987.00
2026-02-10	Evacuation (no fire)	HARINGEY	236	1837.00
2026-03-18	Assist other agencies	CROYDON	236	1837.00
2026-04-15	RTC	BEXLEY	235	1970.00
2026-03-25	Hazardous Materials incident	HAMMERSMITH AND FULHAM	234	1821.00
2026-05-30	Hazardous	BRENT	234	1962.00

date_of_call	special_service_type	borough_name	pump_minutes_rounded	notional_cost_gbp
	Materials incident			
2026-01-09	Hazardous Materials incident	EALING	233	1814.00
2026-03-11	Hazardous Materials incident	WESTMINSTER	232	1806.00
2026-04-17	RTC	ISLINGTON	232	1945.00
2026-04-13	Hazardous Materials incident	HILLINGDON	230	1928.00
2026-02-27	RTC	HILLINGDON	229	1782.00
2026-06-24	RTC	NEWHAM	229	1920.00

Showing 20 of 858 rows.

Interpretation: 858 special services kept engines busy for 2–4 hours, a varied mix led by hazardous materials, road traffic collisions and flooding, each costing roughly £1,000–£2,000.

Q50

Question: Which fires in properties whose type mentions "flat" cost more than £5,000? Show the date, borough, ward, property type, number of pumps attending and notional cost, highest cost first.

Concept: Basic WHERE - LIKE

date_of_call	borough_name	ward_name	property_type	num_pumps_attending	notional_cost_gbp
2025-08-20	LEWISHAM	NEW CROSS GATE	Purpose Built Flats/Maisonettes - Up to 3 storeys	2	201612.00
2026-04-30	HACKNEY	CAZENOVE	Converted Flat/Maisonettes - 3 or more storeys	4	65021.00
2025-05-15	WALTHAM FOREST	ST. JAMES	Purpose Built Flats/Maisonettes - 4 to 9 storeys	4	63769.00
2025-09-10	WANDSWORT	FALCONBROO	Purpose Built	5	56040.00

date_of_call	borough_name	ward_name	property_type	num_pumps_attending	notional_cost_gbp
	H	K	Flats/Maisonettes - 4 to 9 storeys		
2026-05-20	WESTMINSTER	MAIDA VALE	Purpose Built Flats/Maisonettes - 10 or more storeys	5	53570.00
2025-07-08	LEWISHAM	LEWISHAM CENTRAL	Purpose Built Flats/Maisonettes - 4 to 9 storeys	6	47611.00
2026-05-05	HACKNEY	SPRINGFIELD	Purpose Built Flats/Maisonettes - 4 to 9 storeys	6	46318.00
2025-04-20	TOWER HAMLETS	WHITECHAPEL	Purpose Built Flats/Maisonettes - 10 or more storeys	5	46093.00
2025-11-01	KENSINGTON AND CHELSEA	ST. HELEN'S	Converted Flat/Maisonettes - 3 or more storeys	3	45976.00
2025-09-22	ENFIELD	CARTERHATCH	Purpose Built Flats/Maisonettes - Up to 3 storeys	4	41983.00
2025-06-19	HAMMERSMITH AND FULHAM	COLLEGE PARK & OLD OAK	Purpose Built Flats/Maisonettes - Up to 3 storeys	4	40754.00
2025-07-24	BEXLEY	LONGLANDS	Purpose Built Flats/Maisonettes - Up to 3 storeys	4	40364.00
2025-03-18	CAMDEN	BELSIZE	Converted	6	38514.00

date_of_call	borough_name	ward_name	property_type	num_pumps_attending	notional_cost_gbp
			Flat/Maisonettes - 3 or more storeys		
2026-04-08	EALING	PERIVALE	Purpose Built Flats/Maisonettes - Up to 3 storeys	3	35336.00
2025-11-11	HILLINGDON	NORTHWOOD HILLS	Purpose Built Flats/Maisonettes - Up to 3 storeys	2	34433.00
2025-12-19	EALING	EALING COMMON	Converted Flat/Maisonettes - 3 or more storeys	2	33336.00
2026-05-11	WANDSWORTH	BATTERSEA PARK	Purpose Built Flats/Maisonettes - 10 or more storeys	5	32058.00
2025-05-14	LAMBETH	STREATHAM ST. LEONARD'S	Converted Flat/Maisonettes - 3 or more storeys	2	30713.00
2025-04-09	TOWER HAMLETS	BETHNAL GREEN	Purpose Built Flats/Maisonettes - 4 to 9 storeys	5	30215.00
2026-03-14	WALTHAM FOREST	VALLEY	Converted Flat/Maisonette - Up to 2 storeys	2	29553.00

Showing 20 of 219 rows.

Interpretation: Fires in flats can be very costly, with one Lewisham fire reaching £201,612, and fires in high-rise flats often need five or more pumps.

Q51

Question: How is the brigade's notional cost split across incident groups? For each of fires, special services and false alarms, show the number of incidents, total cost, average cost per incident and total pump minutes.

Concept: GROUP BY / aggregation

incident_group	incidents	total_cost	avg_cost	total_pump_minutes
False Alarm	93736	46183215.00	492.69	5924728
Special Service	86851	45509573.00	524.00	5836354
Fire	29183	32768936.00	1122.88	4189460
NULL	138	71370.00	517.17	8858

4 rows returned (shown in full).

Interpretation: False alarms cost the most in total (£46.2m), almost the same as special services (£45.5m). Fires cost less overall (£32.8m) but have the highest average cost (£1,123), so cutting false alarms would save the most money.

Q52

Question: Which 15 wards account for the most fire engine time? For each ward, show its borough, the number of incidents, total pump minutes, total notional cost and average pump minutes per incident, highest total pump minutes first.

Concept: GROUP BY / aggregation, LIMIT

ward_name	borough_name	incidents	total_pump_minutes	total_cost	avg_pump_minutes
WEST END	WESTMINSTER	2596	179947	1401393.00	69.3
ST. JAMES'S	WESTMINSTER	2024	145010	1132040.00	71.6
SLADE GREEN & NORTHEND	BEXLEY	297	113685	895028.00	382.8
HEATHROW VILLAGES	HILLINGDON	1246	105540	822140.00	84.7
PINKWELL	HILLINGDON	351	94943	697141.00	270.5
BLOOMSBURY	CAMDEN	1219	84261	654950.00	69.1
DOLLIS HILL	BRENT	510	77678	631534.00	152.3
HOLBORN &	CAMDEN	964	75156	589696.00	78.0

ward_name	borough_name	incidents	total_pump_minutes	total_cost	avg_pump_minutes
COVENT GARDEN					
MARYLEBONE	WESTMINSTER	778	74623	572424.00	95.9
NORWOOD GREEN	EALING	337	70207	546171.00	208.3
RAINHAM & WENNINGTON	HAVERING	563	66855	523743.00	118.7
WHITECHAPEL	TOWER HAMLETS	855	66506	518961.00	77.8
CHURCH STREET	WESTMINSTER	844	65502	509186.00	77.6
FAIRFIELD	CROYDON	887	63400	495799.00	71.5
NEW CROSS GATE	LEWISHAM	259	62146	493620.00	239.9

15 rows returned (shown in full).

Interpretation: West End and St. James's use the most engine time overall, but Slade Green & Northend in Bexley averages 382.8 pump minutes per incident, so a few large incidents there take up a lot of resource.

Q53

Question: How much do false alarms cost in each property category? Considering only false alarms, show the number of incidents, total cost and average cost per category. Include only categories where false alarms cost more than £100,000 in total, highest first.

Concept: GROUP BY / aggregation with HAVING

property_category	false_alarms	total_cost	avg_cost
Dwelling	60442	30076569.00	497.61
Non Residential	20116	9731166.00	483.75
Other Residential	7951	3809989.00	479.18
Outdoor	2955	1445259.00	489.09
Road Vehicle	1111	534413.00	481.02
Outdoor Structure	1022	496701.00	486.01

6 rows returned (shown in full).

Interpretation: Dwellings account for the largest false alarm cost (£30.1m), and the average cost is about £480–£500 in every category, so the problem is the volume of false alarms rather than the cost of each.

Q54

Question: How does cost rise as more engines attend? For each number of pumps attending, show the number of incidents, the average notional cost and the average pump minutes, in order of pumps attending. Exclude incidents with no number of pumps recorded.

Concept: GROUP BY / aggregation

num_pumps_attending	incidents	avg_cost	avg_pump_minutes
1	126845	497.17	63.7
2	62755	616.48	79.1
3	6994	1045.36	134.9
4	3099	1847.57	235.5
5	7427	973.55	124.4
6	385	2537.75	325.4
7	54	6170.52	789.1
8	11	3645.91	460.3
9	2	9962.50	1306.5
10	2	7573.00	960.0

10 rows returned (shown in full).

Interpretation: Cost generally rises as more engines attend, but 5-pump incidents are cheaper than 4-pump ones, likely because many automatic alarms receive five pumps but finish quickly.

Q55

Question: For each special service type, how many fire engines were sent and how long did they spend on scene on average, from arriving to leaving? Show the time in minutes, excluding engines with no recorded arrival or leaving time, longest average first.

Concept: Join (INNER JOIN) with aggregation

special_service_type	engines_sent	avg_minutes_on_scene
NULL	21	57.0

special_service_type	engines_sent	avg_minutes_on_scene
Evacuation (no fire)	351	54.5
Hazardous Materials incident	3046	52.5
Rescue or evacuation from water	401	45.0
Stand By	282	42.5
Other rescue/release of persons	787	40.2
Making Safe (not RTC)	1815	38.3
Suicide/attempts	1287	38.1
Medical Incident	1726	30.8
RTC	8660	30.1
Other Transport incident	608	29.6
Assist other agencies	8389	29.1
Flooding	13232	27.0
Animal assistance incidents	2157	26.6
Removal of objects from people	1123	25.4
Spills and Leaks (not RTC)	863	24.5
Water provision	5	22.2
Effecting entry/exit	21739	20.9
Advice Only	2290	19.3
Lift Release	4485	16.8
No action (not false alarm)	11386	12.8

21 rows returned (shown in full).

Interpretation: Evacuations and hazardous materials keep engines on scene longest (over 50 minutes), while effecting entry is the most common (21,739 engines) but much quicker at about 21 minutes.

Q56

Question: Within each borough, what were the five most expensive incidents? Rank incidents by notional cost inside their own borough and show the borough, rank, incident number, date, incident group, property type and cost.

Concept: Window function - ranking within a partition

borough_name	cost_rank	incident_number	date_of_call	incident_group	property_type	notional_cost_gbp
BARKING AND DAGENHAM	1	130661-14072025	2025-07-14	Fire	Heathland	56281.00
BARKING AND DAGENHAM	2	061494-07042026	2026-04-07	Fire	Outdoor storage	36166.00
BARKING AND DAGENHAM	3	086327-11052026	2026-05-11	Fire	Unlicensed House in Multiple Occupation - 3 or more storeys	14210.00
BARKING AND DAGENHAM	4	079911-02052026	2026-05-02	Fire	Warehouse	14118.00
BARKING AND DAGENHAM	5	045977-19032025	2025-03-19	Fire	Restaurant/ cafe	12642.00
BARNET	1	098138-27052026	2026-05-27	Fire	Large supermarket	222728.00
BARNET	2	092182-26052025	2025-05-26	Fire	House - single occupancy	85624.00
BARNET	3	022288-04022026	2026-02-04	Fire	House - single occupancy	36263.00
BARNET	4	160387-25082025	2025-08-25	Fire	House - single occupancy	29032.00
BARNET	5	036155-03032025	2025-03-03	Fire	Restaurant/ cafe	28989.00
BEXLEY	1	128104-11072025	2025-07-11	Fire	Recycling plant	447472.00
BEXLEY	2	113689-	2026-06-17	Fire	Recycling	145937.00

borough_name	cost_rank	incident_number	date_of_call	incident_group	property_type	notional_cost_gbp
		17062026			plant	
BEXLEY	3	049640-20032026	2026-03-20	Fire	Warehouse	71941.00
BEXLEY	4	054944-28032026	2026-03-28	Fire	Landfill site	70626.00
BEXLEY	5	097948-04062025	2025-06-04	Fire	Recycling plant	60990.00
BRENT	1	109816-11062026	2026-06-11	Fire	Hairdresser	243712.00
BRENT	2	090799-24052025	2025-05-24	Fire	House - single occupancy	110749.00
BRENT	3	097898-26052026	2026-05-26	Fire	Warehouse	93290.00
BRENT	4	128548-12072025	2025-07-12	Fire	Restaurant/ cafe	46140.00
BRENT	5	116864-21062026	2026-06-21	Fire	Restaurant/ cafe	36652.00

Showing 20 of 165 rows.

Interpretation: In nearly every borough the most expensive incident is a fire, mostly at commercial sites such as warehouses, shops and recycling plants, while Haringey's most expensive incident cost only £10,866.

Q57

Question: Using the daily summary, how did the brigade's total notional cost build up month by month across all 18 months? Show each month's total cost, the running cumulative total, and the change from the previous month in pounds and percentage.

Concept: VIEW (reuses daily summary) + window function (running total, LAG)

month	total_cost	cumulative_cost	change_gbp	change_pct
2025-01	5800435.00	5800435.00	NULL	NULL
2025-02	5065355.00	10865790.00	-735080.00	-12.67
2025-03	6242216.00	17108006.00	1176861.00	23.23

month	total_cost	cumulative_cost	change_gbp	change_pct
2025-04	6541810.00	23649816.00	299594.00	4.80
2025-05	7265867.00	30915683.00	724057.00	11.07
2025-06	7269885.00	38185568.00	4018.00	0.06
2025-07	8223545.00	46409113.00	953660.00	13.12
2025-08	7256420.00	53665533.00	-967125.00	-11.76
2025-09	6743887.00	60409420.00	-512533.00	-7.06
2025-10	6811204.00	67220624.00	67317.00	1.00
2025-11	6835202.00	74055826.00	23998.00	0.35
2025-12	6430178.00	80486004.00	-405024.00	-5.93
2026-01	6918971.00	87404975.00	488793.00	7.60
2026-02	5529785.00	92934760.00	-1389186.00	-20.08
2026-03	6627064.00	99561824.00	1097279.00	19.84
2026-04	7436061.00	106997885.00	808997.00	12.21
2026-05	9064918.00	116062803.00	1628857.00	21.90
2026-06	8470291.00	124533094.00	-594627.00	-6.56

18 rows returned (shown in full).

Interpretation: Total cost reached £124.5m over 18 months. Costs dip each February and peaked in May 2026 (£9.06m), following the same pattern as incident numbers.

Q58

Question: Using the borough summary, which boroughs have an average cost per incident above London's overall average, meaning total cost divided by the number of incidents with a recorded cost? Show each borough's average, London's average and the difference, highest first.

Concept: VIEW (reuses borough summary) + subquery

borough_name	borough_avg_cost	london_avg_cost	difference
BEXLEY	790.75	593.27	197.48
BRENT	686.57	593.27	93.30
HILLINGDON	682.56	593.27	89.28
ENFIELD	666.97	593.27	73.69

borough_name	borough_avg_cost	london_avg_cost	difference
EALING	666.43	593.27	73.15
HAVERING	633.60	593.27	40.33
LEWISHAM	624.09	593.27	30.82
REDBRIDGE	611.50	593.27	18.22
HAMMERSMITH AND FULHAM	611.38	593.27	18.11
BARNET	606.42	593.27	13.15
KINGSTON UPON THAMES	604.80	593.27	11.52
NEWHAM	604.76	593.27	11.49
HOUNSLOW	601.35	593.27	8.07
HARROW	601.27	593.27	8.00
BARKING AND DAGENHAM	598.15	593.27	4.87

15 rows returned (shown in full).

Interpretation: 15 boroughs are above London's average cost per incident (£593.27), mostly outer boroughs, with Bexley far ahead (£790.75).

Q59

Question: Do incidents where at least one engine had an actual delay cost more than incidents with no delay? Classify each incident that had at least one engine mobilised as delayed or not delayed, then compare the number of incidents, average notional cost and average pump minutes for the two groups.

Concept: CTE + join

delay_status	incidents	avg_cost	avg_pump_minutes
Not delayed	169994	592.45	76.0
Delayed	28613	637.07	81.3

2 rows returned (shown in full).

Interpretation: Delayed incidents cost more on average (£637 against £592) and use more pump minutes, so reducing delays could also reduce costs.

Q60

Question: Using the station ground summary, which station grounds have both a false alarm percentage above London's overall false alarm percentage and a total notional cost above the average station ground? Show their false alarm percentage and total cost, highest cost first, as the places where reducing false alarms would free the most resource.

Concept: VIEW (reuses station ground summary) + subquery

incident_station_ground	false_alarm_pct	total_cost
Paddington	53.04	2945843.00
Soho	61.00	2763697.00
Lambeth	45.28	2337307.00
Euston	56.20	2296552.00
North Kensington	51.38	2270127.00
Hammersmith	53.04	2132585.00
West Hampstead	47.27	2130317.00
Shoreditch	47.81	1898532.00
Chelsea	60.19	1780676.00
Kentish Town	48.60	1700212.00
Brixton	47.75	1555071.00
Wandsworth	48.09	1413744.00
Heston	46.63	1350158.00
Peckham	45.85	1323588.00
Tooting	50.76	1319618.00

15 rows returned (shown in full).

Interpretation: 15 station grounds combine high false alarm rates with high costs, led by Paddington (£2.9m) and Soho (61% false alarms), making them the best places to target false alarm reduction.

5.5 Perspective 5: What types of incidents occur and their impact

Q61

Question: What types of special service does the brigade attend? List each special service type once, excluding incidents with no type recorded, in alphabetical order.

Concept: Basic SELECT - DISTINCT

special_service_type
Advice Only
Animal assistance incidents
Assist other agencies
Effecting entry/exit
Evacuation (no fire)
Flooding
Hazardous Materials incident
Lift Release
Making Safe (not RTC)
Medical Incident
No action (not false alarm)
Other rescue/release of persons
Other Transport incident
Removal of objects from people
Rescue or evacuation from water
RTC
Spills and Leaks (not RTC)
Stand By
Suicide/attempts
Water provision

20 rows returned (shown in full).

Interpretation: There are 20 types of special service, from flooding and lift releases to hazardous materials and water rescues, showing how varied the brigade's non-fire work is.

Q62

Question: Which fires prompted 10 or more calls from the public? Show the date, time of call, borough, property type, number of calls and number of pumps attending, most calls first.

Concept: Basic WHERE - comparison

date_of_call	time_of_call	borough_name	property_type	num_calls	num_pumps_attending
2025-03-20	23:23:34	HILLINGDON	Other outdoor equipment/machinery	210	3
2025-04-29	05:29:38	WESTMINSTER	Other outdoor equipment/machinery	174	2
2026-06-08	17:32:13	LEWISHAM	Recycling plant	153	5
2026-06-11	21:14:34	BRENT	Hairdresser	93	4
2025-07-14	18:31:29	BARKING AND DAGENHAM	Heathland	86	2
2025-08-03	21:48:05	RICHMOND UPON THAMES	Tree scrub	85	6
2025-11-25	08:55:13	EALING	Other retail	79	4
2026-06-24	18:23:39	WESTMINSTER	Bus/coach	78	4
2025-07-11	13:55:05	SOUTHWARK	Purpose Built Flats/Maisonettes - 10 or more storeys	75	6
2025-07-12	14:42:06	BRENT	Restaurant/cafe	72	4
2026-05-19	16:23:38	KENSINGTON AND CHELSEA	Outdoor storage	71	4
2026-05-24	18:10:21	HACKNEY	Towing caravan (not on tow or on site)	71	1
2025-05-15	15:46:55	NEWHAM	Private Garden Shed	67	4
2026-04-05	17:16:05	WALTHAM FOREST	Railway trackside vegetation	66	2

date_of_call	time_of_call	borough_name	property_type	num_calls	num_pumps_attending
2026-05-05	20:04:31	HACKNEY	Purpose Built Flats/Maisonettes - 4 to 9 storeys	66	6
2026-04-30	16:29:42	HAVERING	Roadside vegetation	64	2
2026-03-22	11:46:10	WALTHAM FOREST	Refuse/rubbish tip	63	3
2026-05-27	06:47:34	BARNET	Large supermarket	63	2
2026-06-03	20:47:32	WALTHAM FOREST	Private Summer house	63	6
2026-03-23	01:40:53	BARNET	Multiple Vehicles	61	2

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: The fires with the most calls are large, visible outdoor or commercial fires, with one in Hillingdon receiving 210 calls, yet some needed only one or two pumps, so call volume does not always match the size of the response.

Q63

Question: Which incidents needed engines from four or more different stations? Show the incident number, date, borough, incident group, property type, number of stations with pumps attending and number of pumps attending, largest first.

Concept: Basic WHERE - comparison

incident_number	date_of_call	borough_name	incident_group	property_type	num_stations_with_pumps_attending	num_pumps_attending
006729-11012025	2025-01-11	BARKING AND DAGENHAM	Fire	Outdoor storage	9	9
066143-14042026	2026-04-14	HACKNEY	Fire	Purpose Built Flats/Maiso	8	10

incident_number	date_of_call	borough_name	incident_group	property_type	num_stations_with_pumps_attending	num_pumps_attending
				nettes - 10 or more storeys		
093130-27052025	2025-05-27	TOWER HAMLETS	False Alarm	Purpose Built Flats/Maisonettes - 10 or more storeys	8	8
094865-30052025	2025-05-30	BARKING AND DAGENHAM	Fire	Recycling plant	8	8
036598-27022026	2026-02-27	HACKNEY	Fire	Purpose Built Flats/Maisonettes - 10 or more storeys	7	9
017262-27012026	2026-01-27	ISLINGTON	False Alarm	Purpose Built Flats/Maisonettes - 4 to 9 storeys	7	8
034173-27022025	2025-02-27	LAMBETH	False Alarm	Converted Flat/Maisonettes - 3 or more storeys	7	8
095064-30052025	2025-05-30	WANDSWORTH	Fire	Purpose Built Flats/Maisonettes - 4 to 9 storeys	7	8
056433-30032026	2026-03-30	TOWER HAMLETS	Fire	Purpose Built	7	8

incident_number	date_of_call	borough_name	incident_group	property_type	num_stations_with_pumps_attending	num_pumps_attending
				Flats/Maisonettes - 4 to 9 storeys		
052669-29032025	2025-03-29	NEWHAM	Fire	Purpose Built Flats/Maisonettes - 4 to 9 storeys	7	7
087486-13052026	2026-05-13	TOWER HAMLETS	Fire	Purpose Built Flats/Maisonettes - 10 or more storeys	7	7
010944-17012025	2025-01-17	SOUTHWARK	False Alarm	Purpose Built Flats/Maisonettes - 10 or more storeys	7	7
029592-20022025	2025-02-20	SOUTHWARK	False Alarm	Hotel/motel	7	7
174545-16092025	2025-09-16	LAMBETH	Fire	Converted Flat/Maisonettes - 3 or more storeys	7	7
191404-11102025	2025-10-11	SOUTHWARK	Fire	Other building/use not known	7	7
195020-16102025	2025-10-16	LEWISHAM	Fire	Single shop	7	7
112077-23062025	2025-06-23	SOUTHWARK	Fire	Purpose Built Flats/Maiso	7	7

incident_number	date_of_call	borough_name	incident_group	property_type	num_stations_with_pumps_attending	num_pumps_attending
				nettes - 10 or more storeys		
049082-23032025	2025-03-23	WANDSWORTH	Fire	Purpose Built Flats/Maisonettes - 4 to 9 storeys	6	8
076878-29042026	2026-04-29	BARKING AND DAGENHAM	Fire	Purpose Built Flats/Maisonettes - 10 or more storeys	6	8
149887-11082025	2025-08-11	ENFIELD	Fire	Purpose Built Flats/Maisonettes - 10 or more storeys	6	8

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: Most incidents needing four or more stations are in high-rise flats, but a notable number are false alarms, especially in Tower Hamlets and Southwark, which is a large use of resource.

Q64

Question: Which special service incidents had a type that mentions flooding? Show the date, borough, ward, property type, special service type and pump minutes, longest pump time first.

Concept: Basic WHERE - LIKE

date_of_call	borough_name	ward_name	property_type	special_service_type	pump_minutes_rounded
2025-09-25	WESTMINSTER	MARYLEBONE	Road surface/pavement	Flooding	4904
2025-08-10	ISLINGTON	CANONBURY	Pipe or drain	Flooding	3376

date_of_call	borough_name	ward_name	property_type	special_service_type	pump_minutes_rounded
2025-02-27	LEWISHAM	LEWISHAM CENTRAL	Purpose built office	Flooding	3058
2026-02-24	ISLINGTON	CALEDONIAN	Road surface/pavement	Flooding	2515
2026-01-21	KENSINGTON AND CHELSEA	NORLAND	Other outdoor structures	Flooding	1819
2026-05-03	RICHMOND UPON THAMES	HAM, PETERSHAM & RICHMOND RIVERSIDE	Road surface/pavement	Flooding	1706
2025-10-31	CAMDEN	KENTISH TOWN SOUTH	Pipe or drain	Flooding	1326
2026-04-25	BARNET	HENDON	Pipe or drain	Flooding	1116
2026-01-04	ISLINGTON	FINSBURY PARK	Converted Flat/Maisonettes - 3 or more storeys	Flooding	1080
2025-01-21	SOUTHWARK	PECKHAM RYE	Road surface/pavement	Flooding	1057
2026-01-12	KENSINGTON AND CHELSEA	STANLEY	Purpose Built Flats/Maisonettes - Up to 3 storeys	Flooding	998
2025-02-16	WESTMINSTER	PIMLICO SOUTH	Road surface/pavement	Flooding	903
2026-05-28	SOUTHWARK	RYE LANE	Road surface/pavement	Flooding	813
2025-04-01	ISLINGTON	CALEDONIAN	Takeaway/ fast food	Flooding	758
2025-02-14	BEXLEY	ST. MARY'S &	Road surface/pavem	Flooding	726

date_of_call	borough_name	ward_name	property_type	special_service_type	pump_minutes_rounded
		ST. JAMES	ent		
2025-06-01	HILLINGDON	YIEWSLEY	Road surface/pavement	Flooding	713
2026-04-10	HARINGEY	NOEL PARK	Purpose Built Flats/Maisonettes - 10 or more storeys	Flooding	697
2025-05-15	BRENT	WEMBLEY PARK	Purpose Built Flats/Maisonettes - 10 or more storeys	Flooding	673
2026-04-13	HARINGEY	STROUD GREEN	Purpose Built Flats/Maisonettes - 4 to 9 storeys	Flooding	667
2026-06-23	HILLINGDON	HEATHROW VILLAGES	Warehouse	Flooding	627

Showing 20 of 1,000 rows (export capped at 1,000).

Interpretation: Flooding incidents can use a lot of pump time, up to 4,904 minutes in Marylebone, mostly on roads, drains and in flats, so flooding is a bigger demand than expected.

Q65

Question: Which incidents were recorded as malicious false alarms between January and June 2026? Show the date, time of call, borough, ward and property type, most recent first.

Concept: Basic WHERE - LIKE, BETWEEN

date_of_call	time_of_call	borough_name	ward_name	property_type
2026-06-29	17:59:21	SOUTHWARK	CAMBERWELL GREEN	Restaurant/cafe
2026-06-29	11:53:50	WESTMINSTER	ST. JAMES'S	Hotel/motel
2026-06-29	11:48:12	LAMBETH	CLAPHAM PARK	Purpose Built Flats/Maisonettes - Up to 3 storeys

date_of_call	time_of_call	borough_name	ward_name	property_type
2026-06-28	21:16:10	SOUTHWARK	NUNHEAD & QUEEN'S ROAD	False Alarm - Property not found
2026-06-28	19:03:51	KENSINGTON AND CHELSEA	REDCLIFFE	Car
2026-06-28	15:17:35	TOWER HAMLETS	BLACKWALL & CUBITT TOWN	Purpose Built Flats/Maisonettes - 4 to 9 storeys
2026-06-28	00:26:26	BARNET	CRICKLEWOOD	House - single occupancy
2026-06-27	16:12:35	BROMLEY	KELSEY & EDEN PARK	Medical/health centre
2026-06-27	13:10:17	REDBRIDGE	VALENTINES	Telephone box
2026-06-27	13:08:48	KINGSTON UPON THAMES	ST. MARK'S & SEETHING WELLS	Car
2026-06-27	04:23:28	LEWISHAM	LADYWELL	Park
2026-06-27	00:28:24	MERTON	WIMBLEDON PARK	Purpose Built Flats/Maisonettes - 4 to 9 storeys
2026-06-26	19:43:44	WALTHAM FOREST	LEA BRIDGE	Human harm outdoors
2026-06-26	18:37:14	NEWHAM	EAST HAM SOUTH	Purpose Built Flats/Maisonettes - 4 to 9 storeys
2026-06-26	18:03:16	HACKNEY	LONDON FIELDS	Converted Flat/Maisonettes - 3 or more storeys
2026-06-26	17:43:10	HILLINGDON	COLHAM & COWLEY	Hospital
2026-06-26	17:16:58	REDBRIDGE	MONKHAMS	House - single occupancy
2026-06-26	16:17:51	KINGSTON UPON THAMES	KINGSTON TOWN	Single shop

date_of_call	time_of_call	borough_name	ward_name	property_type
2026-06-25	23:44:02	HAMMERSMITH AND FULHAM	MUNSTER	Converted Flat/Maisonettes - 3 or more storeys
2026-06-25	12:09:10	LEWISHAM	BELLINGHAM	House - single occupancy

Showing 20 of 531 rows.

Interpretation: There were 531 malicious false alarms in January–June 2026 across all boroughs, with repeat locations such as Clayhall in Redbridge, St. Paul's Cray in Bromley and East Finchley in Barnet, so these areas should be targeted.

Q66

Question: What is the typical impact of each detailed incident category? For each category, show the number of incidents, the average number of pumps attending, the average pump minutes and the average number of calls received, highest average pump minutes first.

Concept: GROUP BY / aggregation

stop_code_description	incidents	avg_pumps	avg_pump_minutes	avg_calls
Primary Fire	13525	2.15	233.9	2.98
Chimney Fire	17	2.00	109.9	1.35
Alleged Fire Risk	176	1.00	69.1	1.04
Special Service	86694	1.10	67.2	1.17
Secondary Fire	15629	1.17	65.5	1.72
False alarm - Good intent	20218	1.77	65.4	1.17
Flood call attended - Batch mobilised	112	1.06	64.5	1.09
False alarm - Malicious	1858	1.71	62.7	1.19
AFA	71651	2.04	62.6	1.18
Late Call	12	1.00	62.1	1.08
Use of Special Operations Room	16	1.00	60.0	1.06

11 rows returned (shown in full).

Interpretation: Primary fires are the most resource-heavy (2.15 pumps, 233.9 pump minutes), while AFAs are very common (71,651) and still send about two pumps each, adding up to a lot of wasted engine time.

Q67

Question: Which special service types most often need more than one fire engine? For each type with at least 500 incidents, show the number of incidents and the percentage where two or more pumps attended, based on incidents with a recorded pump count, highest percentage first.

Concept: GROUP BY / aggregation with HAVING

special_service_type	incidents	pct_two_plus_pumps
Suicide/attempts	845	43.93
RTC	6676	34.93
Hazardous Materials incident	2313	31.71
Other rescue/release of persons	676	19.52
Making Safe (not RTC)	1612	11.70
Medical Incident	1706	10.34
Advice Only	2097	9.91
No action (not false alarm)	11883	9.78
Spills and Leaks (not RTC)	805	9.33
Assist other agencies	8045	6.23
Flooding	12807	3.70
Removal of objects from people	1222	3.50
Animal assistance incidents	2173	2.14
Effecting entry/exit	22066	1.09
Lift Release	10819	0.75

15 rows returned (shown in full).

Interpretation: Suicide attempts (43.9%), road traffic collisions (34.9%) and hazardous materials (31.7%) most often need two or more engines, while lift releases and effecting entry almost always need just one.

Q68

Question: Which 10 property types have the most fires? For each, show the number of fires, the average number of pumps attending and the average pump minutes per fire.

Concept: GROUP BY / aggregation, LIMIT

property_type	fires	avg_pumps	avg_pump_minutes
Small refuse/rubbish container	4048	1.06	60.2
House - single occupancy	2341	1.98	196.9
Domestic garden (vegetation not equipment)	1704	1.31	63.3
Loose refuse	1622	1.29	66.4
Purpose Built Flats/Maisonettes - 4 to 9 storeys	1606	2.85	199.9
Purpose Built Flats/Maisonettes - Up to 3 storeys	1440	2.07	174.7
Tree scrub	1283	1.08	70.0
Car	1198	1.25	82.8
Large refuse/rubbish container (eg skip/ paladin)	928	1.29	69.5
Wheelie bin (domestic size)	755	1.10	60.4

10 rows returned (shown in full).

Interpretation: Small refuse containers have the most fires (4,048) but need little resource, while flats of 4 to 9 storeys need the most pumps (2.85) and time (199.9 minutes), so flats are the bigger risk.

Q69

Question: Which types of incident most often draw fire engines from outside their home station? For each incident group, show the number of engines mobilised, the number not at their home station when mobilised, and that number as a percentage of all engines mobilised.

Concept: Join (INNER JOIN) with aggregation

incident_group	engines_mobilised	not_from_home	pct_not_from_home
Fire	45906	1617	3.52
False Alarm	182208	6183	3.39
Special Service	84685	2801	3.31
NULL	18	0	0.00

4 rows returned (shown in full).

Interpretation: Only 3.3% to 3.5% of engines are mobilised away from their home station, and this is consistent across all incident types.

Q70

Question: Where exactly do fires happen relative to the recorded address? Considering only fires, show for each address qualifier the number of fires, the average number of pumps attending and the average pump minutes, most common first.

Concept: GROUP BY / aggregation

address_qualifier	fires	avg_pumps	avg_pump_minutes
Correct incident location	8101	2.17	230.0
On land associated with building	7451	1.36	96.4
In street outside gazetteer location	4867	1.13	66.6
Within same building	3056	2.45	259.0
Open land/water - nearest gazetteer location	2897	1.14	91.3
In street close to gazetteer location	1985	1.14	67.2
Nearby address - street not listed in gazetteer	208	1.48	92.0
Nearby address - no building in street	207	1.31	127.3
In street remote from gazetteer location	169	1.15	116.3

address_qualifier	fires	avg_pumps	avg_pump_minutes
Railway land or rolling stock	130	1.78	115.7
On motorway / elevated road	112	1.82	123.2

11 rows returned (shown in full).

Interpretation: Most fires happen at the recorded address or on land around the building, but fires elsewhere within the same building need the most pumps (2.45) and time (259 minutes).

Q71

Question: In each borough, what is the single most common type of special service? Show the borough, the special service type, its number of incidents and its share of that borough's special service incidents. If two types tie, show both.

Concept: CTE + window function (RANK, keeps ties)

borough_name	special_service_type	incidents	share_pct
BARKING AND DAGENHAM	Effecting entry/exit	528	27.67
BARNET	Effecting entry/exit	707	23.13
BEXLEY	Effecting entry/exit	431	24.21
BRENT	Effecting entry/exit	824	27.47
BROMLEY	Effecting entry/exit	485	19.58
CAMDEN	Effecting entry/exit	922	27.07
CITY OF LONDON	Lift Release	138	40.12
CROYDON	Effecting entry/exit	1116	28.03
EALING	Effecting entry/exit	690	23.10
ENFIELD	Effecting entry/exit	739	26.03
GREENWICH	Effecting entry/exit	857	28.37
HACKNEY	Effecting entry/exit	1100	27.02
HAMMERSMITH AND FULHAM	Effecting entry/exit	634	29.77
HARINGEY	Effecting entry/exit	706	26.53

borough_name	special_service_type	incidents	share_pct
HARROW	Effecting entry/exit	341	21.30
HAVERING	Effecting entry/exit	386	22.13
HILLINGDON	RTC	470	19.65
HOUNSLOW	Effecting entry/exit	482	24.10
ISLINGTON	Effecting entry/exit	966	29.23
KENSINGTON AND CHELSEA	Effecting entry/exit	567	22.36
KINGSTON UPON THAMES	Effecting entry/exit	198	19.41
LAMBETH	Effecting entry/exit	1009	24.33
LEWISHAM	Effecting entry/exit	978	27.72
MERTON	Effecting entry/exit	384	27.91
NEWHAM	Effecting entry/exit	870	28.52
REDBRIDGE	Effecting entry/exit	518	25.08
RICHMOND UPON THAMES	Effecting entry/exit	278	21.55
SOUTHWARK	Effecting entry/exit	1128	26.43
SUTTON	Effecting entry/exit	342	26.27
TOWER HAMLETS	Effecting entry/exit	1029	23.61
WALTHAM FOREST	Effecting entry/exit	645	30.67
WANDSWORTH	Effecting entry/exit	842	30.04
WESTMINSTER	Lift Release	994	23.36

33 rows returned (shown in full).

Interpretation: Effecting entry is the most common special service in almost every borough. The exceptions are the City of London and Westminster (lift releases) and Hillingdon (road traffic collisions, likely due to Heathrow and major roads).

Q72

Question: How often do incidents in each property category escalate beyond the initial attendance? Classify each incident that had at least one engine mobilised by whether any engine was sent as

additional resources. For each property category, show the number of incidents, the number that escalated and the escalation percentage, and rank the categories from most to least likely to escalate.

Concept: CTE + join

escalation_rank	property_category	incidents	escalated	escalation_pct
1	Dwelling	123764	0	0.00
1	Non Residential	28053	0	0.00
1	Road Vehicle	11187	0	0.00
1	Outdoor Structure	9502	0	0.00
1	Outdoor	15810	0	0.00
1	Other Residential	9820	0	0.00
1	Aircraft	197	0	0.00
1	Rail Vehicle	160	0	0.00
1	Boat	112	0	0.00
1	NULL	2	0	0.00

10 rows returned (shown in full).

Interpretation: No escalations were found because the source field for additional resources holds the same value for every mobilisation (see section 1.6), so escalation cannot be identified from this dataset.

Q73

Question: For incidents where at least three engines arrived, how long after the first engine did the third one arrive? Number the engines within each incident by their recorded arrival time (not the recorded pump order), excluding engines with no arrival time. Then, for each incident group, show the number of such incidents and the average gap in minutes between the first and third arrivals.

Concept: CTE + window function (ROW_NUMBER)

incident_group	incidents	avg_gap_first_to_third_min
Fire	3013	6.24
Special Service	983	3.90
False Alarm	13094	2.78

3 rows returned (shown in full).

Interpretation: For fires, the third engine arrives on average 6.24 minutes after the first, much longer than for false alarms (2.78 minutes), so back-up takes longer to reach the incidents that need it most.

Q74

Question: Which properties suffer repeated false alarms in quick succession? For each property (by property reference number, ignoring incidents with no reference), find the gap in days between consecutive false alarms. List the properties with at least three false alarms that followed the previous one within 30 days, with their borough, property type and number of quick repeats, most repeats first.

Concept: CTE + window function (LAG within a partition)

uprn	borough_name	property_type	quick_repeats
5124020	CAMDEN	Hospital	122
10070703694	HAVERING	Woodland/forest - broadleaf/hardwood	84
46255350	NEWHAM	Hospital	71
10059101565	HILLINGDON	Road Tanker	51
6741884	TOWER HAMLETS	Hotel/motel	42
5155897	CAMDEN	Hospital	41
100023249499	GREENWICH	Secondary school	40
5087826	CAMDEN	Hospital	39
10093393950	LEWISHAM	Purpose built office	36
5199558	CAMDEN	Student Hall of Residence	34
10090502241	REDBRIDGE	Other Residential Home	30
6053013	TOWER HAMLETS	Single shop	30
100023416227	HILLINGDON	College/University	28
10010439479	ISLINGTON	Hostel (e.g. for homeless people)	27
5133289	CAMDEN	Medical/health centre	27
100023551737	HARINGEY	Single shop	25
10012837041	NEWHAM	Other medical establishment (including surgery)	25
10059101545	HILLINGDON	Passenger plane	25

uprn	borough_name	property_type	quick_repeats
10094085846	SOUTHWARK	Secondary school	25
217040297	KENSINGTON AND CHELSEA	Hotel/motel	24

Showing 20 of 557 rows.

Interpretation: 557 properties had repeat false alarms within 30 days, led by a Camden hospital (122) and a Havering woodland (84). Hospitals, schools and student halls are common, so these sites need alarm reviews.

Q75

Question: Which special service types take up a bigger share of the brigade's special service engine time than their share of special service incidents? For each type, show its share of special service incidents, its share of special service pump minutes and the ratio between the two, highest ratio first.

Concept: CTE + window function (SUM OVER)

special_service_type	incident_share_pct	pump_minute_share_pct	ratio
Rescue or evacuation from water	0.19	0.43	2.26
Stand By	0.16	0.29	1.81
Evacuation (no fire)	0.36	0.64	1.78
Hazardous Materials incident	2.66	4.08	1.53
Suicide/attempts	0.97	1.35	1.39
Other rescue/release of persons	0.78	0.98	1.26
Other Transport incident	0.51	0.60	1.18
Making Safe (not RTC)	1.86	2.15	1.16
RTC	7.69	8.67	1.13
Assist other agencies	9.27	9.41	1.02
Flooding	14.75	14.85	1.01

11 rows returned (shown in full).

Interpretation: Water rescues use more than twice their share of special service engine time (ratio 2.26), and stand-bys, evacuations and hazardous materials also take much more time than their numbers suggest.

6. Summary of Findings

Taken together, these findings show a brigade facing rising and shifting demand. Incidents grew in almost every borough and hour of the day into 2026, with special services and false alarms increasing while fires fell. Demand is heavily concentrated in central London, where Westminster and the Soho and Paddington station grounds carry the greatest load, yet it is the outer boroughs that struggle most to meet the 6- and 8-minute attendance targets.

False alarms are the clearest opportunity for improvement: they are the most common call-out and the most costly in total (£46.2m), with repeat sites such as hospitals, schools and student halls. Response times are slowly getting worse, with traffic the most frequent cause of delay and wrong addresses adding the most time. Flats, especially high-rise blocks, remain the biggest fire risk, and clear peaks in early summer and on Bonfire Night call for planned extra cover.

Finally, missing return times, capped attendance values and incomplete postcodes show that better data recording is needed before some conclusions can be fully trusted.

7. Closing Reflection

This project has been the largest and most difficult SQL task I have undertaken so far. Importing a large dataset was new to me and took a while, although using AI to work through the import problems made it more efficient. The hardest part was writing the queries: each question took time to think through, working out what it was asking and which techniques it needed. The concept stated for each question helped me narrow down the right approach. Exporting the outputs and writing the interpretations were simpler but time-consuming, and AI helped speed up the report by tidying its formatting and wording.

The insight that surprised me most was the sheer number of call-outs, and how many of them were false alarms (93,736 in 18 months, costing £46.2m). Day to day, we rarely hear about fires in the news unless they are large, so I did not expect the brigade's workload to be this high, or for so much of it to be spent on calls that turn out not to be real.